

2026 Baja SAE Design Review Briefing (DRB)

School Name

University of Texas at Austin

2026 Car Number(s)

185

OVERALL VEHICLE PRESENTATION

Design Philosophy

1. Rule Compliance

- a. Design to Margin, leaving $>1"$ & $>3^\circ$ tolerance as per Baja SAE rules
- b. Incorporate 2+ mock tech reviews into our design and fabrication process, ensuring compliance with Baja SAE rules

2. Manufacturability

- a. Utilize OEM components & subsystems where possible, with specific focus on powertrain, and ergonomics
- b. Minimize custom CNC components
- c. Modular Manufacturing to accommodate a non-permanent workspace

3. Durability

- a. Conduct in-depth analysis of components, aiming for a minimum FOS >2 on all critical components for 5-7g's of load
- b. Prioritize durability over weight, ensuring additional supports are added as needed.

4. Serviceability

- a. Critical outboard components replaceable and easily accessible
- b. Minimize variety of fastener sizes
- c. Maximize use of quick fasteners

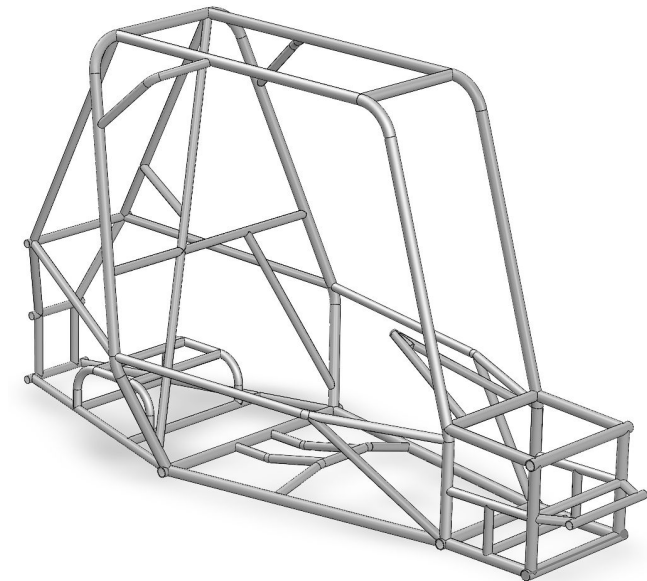
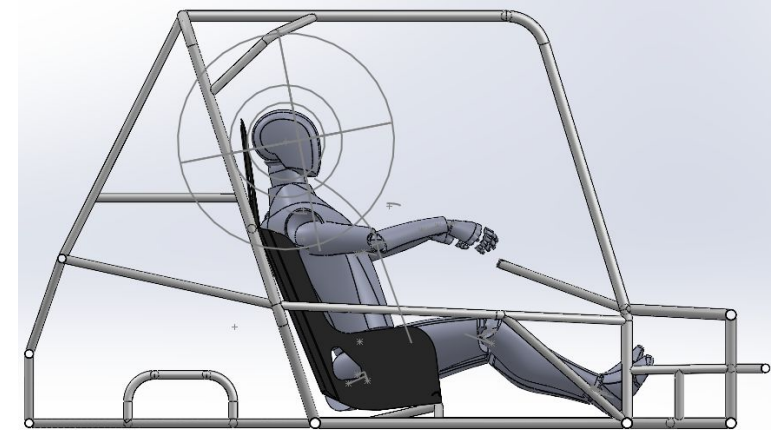
OVERALL VEHICLE PRESENTATION – GOALS AND REQUIREMENTS

Requirement	Design Target / Goal	Why this target was chosen	Result	Notes: (analysis, testing, additional details)
Curb Weight	450 lbs	This target was chosen off of a reasonable estimate for what a car of the Baja SAE type should weigh	N/A	At time of DRB completion the car has yet to achieve powered run, information will be updated as more information is found during testing after all components are on the car.
Weight Bias Front / Rear	40/60	Rear-biased weight bias allows for more rear traction, letting the drive wheels grip better and accelerating the car faster.	40/60	
Wheelbase	72.5 inches	Aimed for a balance of wheelbase length to prevent a worry of rolling risk, and was decided after talking to veteran teams	72.5	
Overall Width (Outside edge of widest rims)	62 inches	Wide track width creates stable handling and lower body roll. 62 chosen for margin of error in manufacturing.	62 inches	
Overall Height (ground to top of roll cage)	67 inches	Was chosen to give chassis and other subteams margin of error room for falling under SAE rules for driver egress from chassis.	N/A	Full Measurement hasn't been performed due to testing with ride height based on shock tuning. Expected ride height for competition will be 67 inches however.
Ground Clearance	12.5 inches	12.5" ground clearance allows for space to clear obstacles without issue		
Tire Size	23x7	Wider tires allow for solid traction over all surfaces and have more capability in supporting the cars weight.	23x7	
Front Suspension Type	Double Wishbone	High kinematic control over camber, caster, and toe allowing for better handling characteristics	Double Wishbone	Contoured upper control arm allowed for shock placement to the lower arm, spreading load to a stronger member.
Front Suspension Travel	12 Inches	12" suspension travel allows the car to handle rough terrain while keeping wheels on the ground, increasing performance	12 inches	Limit straps needed to prevent clearance issue with front uprights. Ideal camber and toe control over full travel create predictable and favorable steering.
Rear Suspension Type	Semi- Trailing Arm	Semi-trailing arm allows for simple construction and ample kinematic control over large suspension travel	Semi- Trailing Arm	Trailing arm with two rear toe links create simple design.
Rear Suspension Travel	12 inches	12" suspension travel allows the car to handle rough terrain while keeping wheels on the ground, increasing performance	12 inches	CV axle plunge and recession may prevent the trailing arm from reaching full 4" droop, constituting the need for limit straps in the rear. Toe control excellent within ideal operating range of suspension travel.
Steering Wheel Rotations lock to lock	1.5	Narrco rack spec for lock to lock. Chosen for availability and proven reputation.	1.5	
Outside Turn Radius	8 feet	Aggressive turning radius allows for tight cornering and more responsive steering feel		
Top Speed	28 MPH	This target was chosen off of initial powertrain estimates.	N/A	At time of DRB completion the car has yet to achieve powered run, information will be updated as more information is found during testing.
Acceleration Time (100ft, 150ft)	N/A	As a first year team initial estimates were hard to get so an initial design target and goal was not set for this.	N/A	At time of DRB completion the car has yet to achieve powered run, information will be updated as more information is found during testing.
Driveline Type (CVT, Sequential, eCVT)	CVT	Was chosen for reliability and after talking to veteran teams	CVT	
4x4 Type (Chain, Shaft, Hydraulic...)	2WD	We have a 2WD car for simplicity as a first year team	2WD	
Vehicle Life Target (miles/hours)	10 hours testing + Competition	As an estimate off of initial timeline for time for testing before competition		

OVERALL VEHICLE PRESENTATION

Chassis and Ergonomics High Level Summary

- **Goals**
 - a. Pass Tech Inspection as a First Year Team
 - i. *>1" & >3° margin on all roll cage related rules*
 - ii. *> 1.5" margin on all driver clearance related rules*
 - iii. *>1" margin on engine rollover*
 - b. Modular chassis jig system adaptable to temporary fabrication space
 - c. Streamline and Simplify Chassis Manufacturing
 - i. *Divide chassis into 5 modular sections for flexible assembly*
 - ii. *Standardize angles to 1° increments*
 - d. In-house Manufacturing
 - i. *Cut, Jig, and Cope all straight members in house*
 - ii. *Outsource bent members*
 - e. Roll Cage Under 75lb (Final Weight = 71.44 lbs)
- **Analysis, Testing, & Results**
 - a. **Frame**
 - 4130 1.25"OD x 0.065" Primary, 1.00"OD x 0.035" Secondary
 - Analyzed torsional rigidity using Ansys FEA
 - Drivetrain & Ergonomics Support (Tertiary Beams)
 - 4130 .125" plate (Engine Mount)
 - 4130 1.00" OD x 0.035" Tertiary (Gas Tank + Brake + Steering Mounts)
 - **Ergonomics**
 - Custom Solidworks driver models to check ranges of actuation and driver clearances for rule-compliance.
 - Utilize OEM seat, pedal box, and steering to ease manufacturing



OVERALL VEHICLE PRESENTATION

Drivetrain High Level Summary

Key Subsystem Goals

- **Reliability:** Rear-wheel drive configuration optimized for durability and consistent power delivery
- **Manufacturability:** First-year team focused on off-the-shelf components and simple, serviceable interfaces
- **Compact Packaging:** Transverse engine orientation minimizes drivetrain length and frees chassis space
- **Serviceability:** Accessible components and simplified assembly for rapid maintenance at competition

R&D & Analysis

- **CVT Thermal Analysis:** Evaluated heat generation and thermal behavior under representative loading
- **Flow Analysis:** Studied airflow around drivetrain components to support CVT cooling
- **Final Drive:** Performed gear ratio, torque, speed, and load calculations to select final-drive gearing
- **Differential Development:** Evaluated differential options for reliability, packaging, cost, and traction
- **System-Level Analysis:** Checked drivetrain loads and interfaces to ensure compatibility with chassis and powertrain components

Testing & Results

- Bench-tested drivetrain components and interfaces prior to vehicle integration
- Utilized OEM Grizzly 660 Differential, CV Axles, and Hubs for reliability and integration
- Validated CVT engagement, belt alignment, and power transmission
- Evaluated differential operation and axle integration
- Prioritized simple, robust solutions appropriate for a first-year competition team

OVERALL VEHICLE PRESENTATION

Suspension, Steering and Brakes High Level Summary

Suspension

- First-year priority: maximize learning by evaluating multiple suspension archetypes (including a CV-axle-integrated lower link rear layout) while committing to a double A-arm baseline proven simple and robust for a first-year team
- Sized all links, joints, and mounts using a custom-built Vehicle Dynamics Calculator (30 sheets, ~7,600 formulas) to apply verified expected loads and validate part robustness before fabrication
- Tuned Fox Factory Race Series dual-rate shocks, optimizing spring crossover position for progressive damping across Baja terrain
- Validated trailing arm and heim joint life-safety margins via bending/two-force-member analysis and bolt stress calculations, sized against calculator-derived loads

Steering

- Implemented a Narrco rack steering configuration to steering column as a lightweight option designed for racing
- Validated steering geometry and bolts/pins against expected loads using the same verified calculator approach as suspension and FEA simulations
- Ensured secure mounting between steering column and steering wheel using OEM Asuza mounting kit for reliability and ease of integration

Brakes

- Built a full brake system calculator to size rotors, calipers, and master cylinders against SAE Baja pedal-force and lockup requirements, ensuring component robustness under expected loading

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FULL VEHICLE ENGINEERING

Benchmarking and Research

- **How did you achieve the measurable data for your vehicle goals?**

- Due to this being our first year and having no prior knowledge base, we relied on looking at competitor vehicles and off the market vehicles like the Yamaha Grizzly and the Canam to learn and base our knowledge off of for our build.

- **What evaluation/testing was completed?**

- Wheel camber relative to CAD
- Suspension shock and tie rod locations relative to CAD
- Measurements of powertrain components' placement relative to CAD

- **How do your vehicle goals compare to the benchmark data collected?**

- Dynamics and powertrain measurements in reality were mostly consistent with CAD. However, due to being a first year team, we have no prior data to measure off of.

- **What research was conducted to help identify potential options that could achieve the vehicle goals**

- What technologies or designs were available to help achieve the goal
- Competitor teams' previous vehicle designs and performance in comp
- Silasbutdeadly's YouTube channel

- **Options that were considered and what goal(s) they could potentially help achieve:**

Powertrain:

- Front differential, to enhance cornering capabilities
- eCVT, to optimize gear ratios for performance
- 4WD/AWD system, to maximize cornering and traction
- Floor mounted engine to lower vehicle's center of gravity

Dynamics:

- Semi-active suspension to optimize dampening rate for better steering, traction, and consistent ride height
- Double wishbone rear suspension for better tuning.

Chassis:

- 1x0.065" Secondary members for increased chassis robustness at the expense of weight
- 3D printed extrusion jig to fixture suspension hard points

Ergo:

- Adjustable pedals allow drivers to choose personalized settings
- Customized seat fit for vehicle integration and driver preferences
- Custom steering wheels for each driver for better grip and feel

Electronics:

- RPM Hall effect sensors for speed measurement
- Dashboard for driver comms
- Fuel gauge for better pit stop timings
- Accelerometer and Gyroscope for telemetry purposes

FULL VEHICLE ENGINEERING

Component & Subsystem Selection

- **How did you select the components/subsystems used?**

Packaging study:

Full Car Mastersketch including packaging constraints for each subsystem to verify proper packaging of all critical components.

Feasibility study:

Looking into in-house fabrication capability, budget constraints, first-year team skill and experience levels weighed against design complexity for each subsystem to alter designs before manufacturing.

Preliminary analysis:

Dynamics sized using custom-built calculators ensuring that all design decisions were based on first principles and would survive the rigor of the terrain.

Chassis used custom matlab plotting algorithm to decide the ideal material for chassis tubes that worked under SAE rules and guidelines, ensuring an affordable and rule compliant spaceframe.

Powertrain used the Yamaha Grizzly 660 differential for its proven off-the-shelf reliability and straightforward CV axle/hub interface, avoiding the complexity of a custom limited-slip build.

- **What components/subsystems did you ultimately select for your vehicle?**

Chassis: 4130 Steel Alloy Tube Frame with 1/8" 4130 Steel Plate Tabs

Dynamics: Double A-arm front, Semi-trailing arm rear, Fox Factory Race Series dual-rate shocks, Narrco Rack, Wilwood GP200 Calipers

Drivetrain: Yamaha 660 Grizzly Differential, CV Axle Splines, Hubs, CV Tech CVT type, gearing, final drive

Electrical: Wiring harness approach, Kill switch, Data logging

Manufacturing: Had access to the university's machine shop, machined parts for each subsystem using CNC, mill, lathe

Simulation & Validation: Ansys and SolidWorks FEA

FULL VEHICLE ENGINEERING

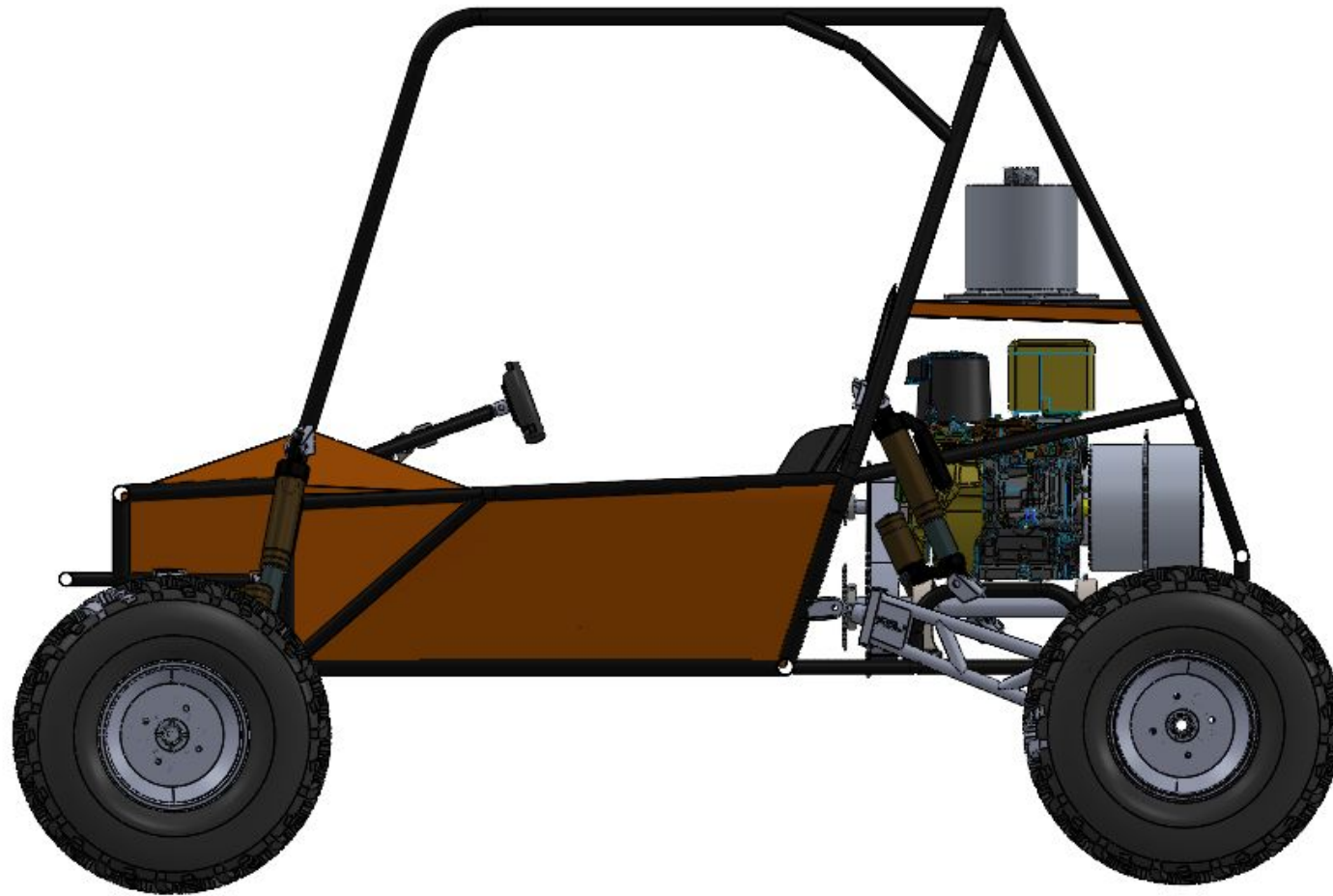
- Top CAD View



Top View

FULL VEHICLE ENGINEERING

Left CAD View



Left Side View

FULL VEHICLE ENGINEERING

Right CAD View



Front View

FULL VEHICLE ENGINEERING

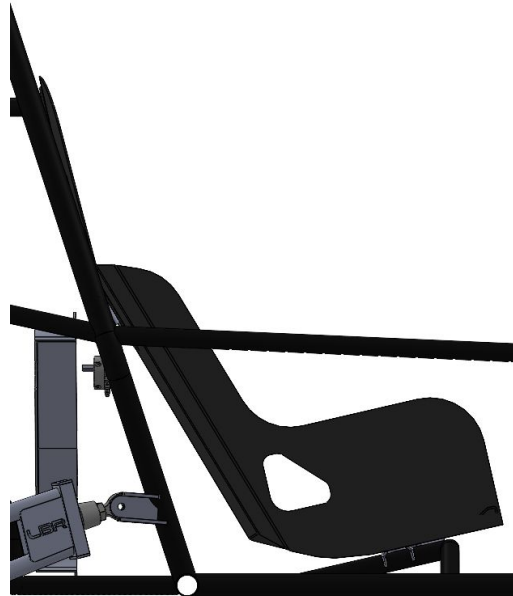
Subsystem Integration



Front Suspension:

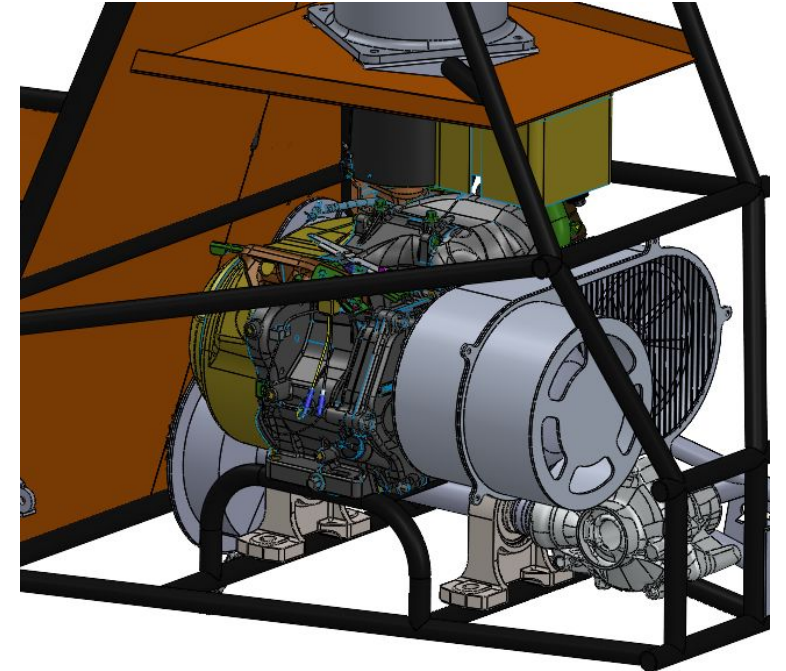
Front control arms at points F_R and F_L in the chassis, providing support for high load stress concentrations.

Ensured rigidity of proper load pathing while sacrificing weight.



Driver Seat Support:

UST angle lifted to accommodate for bucket seat design. Seat altered during manufacturing to optimally meet this angle. UST increases driver security and stability within the cockpit.



Engine Mount

Engine mount was made using Secondary tubing as a hoop to withstand more of the powertrain weight, especially the engine, due to its strong resistance to shear stress. Additionally helps with significantly reducing engine rollover.

FULL VEHICLE ENGINEERING

Full Vehicle / Subsystem Integration Data Collection for Analysis

Suspension Analysis

- Lotus Shark implemented to determine hard points in suspension to accurately simulate vehicle performance.
- Hard points copied to Solidworks and used for integration with chassis and powertrain subsystems, ensuring all components and design were compatible.

Subsystem Clearances

- CAD models made of each component to ensure safe clearances between other subsystem components before implementation in manufacturing
- Ensured gussets on the top of the roll cage would clear the driver's helmet and pass tech by a large margin
- Angle of rear suspension trailing arms was adjust to barely clear tubes in the rear bracing
- Gas tank mount was set with at least 1 inch of clearance above the exhaust
- Clearances of subsystems relative to chassis were generally multiplied by some factor to guarantee clearances were maintained even in the event of manufacturing errors and/or issues during testing/competition.

FEA Boundary Conditions and Loads

- Suspension hard points used as fixed boundary conditions
- Load Cases include 10000 lbf collision forces, 1400 lbf from suspension loads (resulted in additional beams added).
- Loads applied at first-contact collision sites to verify deformation would not affect subsystem clearances, driver safety, and integration.

FULL VEHICLE ENGINEERING

Full Vehicle / Subsystem Integration Testing

- **What physical subsystem or vehicle testing was conducted to demonstrate that your vehicle was capable of meeting your goals?**
 - Length and angle measurements of tubing for in-house notching ensured consistent design between CAD and fabrication
 - Altering purchased seat to optimize fitment
 - Performed interference checks and vehicle fitment with all subsystems installed.
- **What measurable data was collected and how did that compare to the measurable goals you set?**
 - Tube length measurements after cut/notching were within 1/16" from CAD targets (Allowed weld gaps)
 - Driver fitment fitting the 5th-95th percentile driver accommodation
 - Chassis/subsystem clearances: maintained all applicable clearances for driver and powertrain subsystem
- **How did the data that was collected correlate back to your analysis?**
 - Compared measured chassis dimensions to CAD and FEA assumptions
 - Used measured clearance data to validate CAD packaging and general packaging assumptions
- **Demonstrate how the testing results impacted the design of the vehicle. Show test results before and after a design change and explain how the change improved meeting your design goals**
 - Tubing measurements identified some fabrication deviations before welding/tacking which required minor adjustments in weld sizing.
 - Updated CAD and fabrication dimensions to reduce any manufacturing discrepancies.

FULL VEHICLE ENGINEERING

Assessment Of Vehicle Goals

- **Compare the top vehicle goals to what the team actual achieved**
 - Top vehicle goals such as semi active suspension and AWD were not achieved, at the time of DRB submission the team has achieved a DWB Front Suspension and a Semi trailing rear suspension that matches up with simulation estimates as well as a rule compliant chassis. By the time of competition, the team will have achieved a powered car with RWD.
- **What goals did the team succeed at and what goals did the team come up short**
 - The team was able to get RWD car a double wishbone front suspension and a semi-trailing arm rear suspension.
 - The team was also able to achieve a robust rule compliant chassis that weighed less than 80 pounds.
 - Overall the team came short on a lot of reach goals such as semi-active suspension and front wheel drive due to a reprioritization of requirements to ensure a car was able to be completed before competition.
- **For the goals that you came up short on what was the root cause**
 - The biggest root cause for the lack of completion of major team goals was a reprioritization for manufacturability especially as a first year team. A lot of the goals were cut back on to ensure an initial car could be made in a reasonable timeframe giving enough time for setbacks that could occur.

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CHASSIS / ERGONOMICS

Requirements

Compliance:

- Ensure driver safety in event of collision or roll over
- Tubing Meets minimum bending strength & stiffness
- Chassis that meets all roll cage member technical requirements
- Maintain adequate driver clearance and engine roll over clearance

Integration:

- Rear bracing must adequately incorporate drivetrain
- Roll Cage must adequately incorporate driver, steering, and pedal box

Manufacturability:

- Had to ensure the chassis could be assembled in modular sections due to lack of a consistent shop space
- Chassis subsystems must be jigged with $\frac{5}{8}$ " stronghand fixtures
- Straight members must be notchable with a JD2 Tube Notcher

Weight:

- Final chassis weight to be under 75 lbs

Research and Development

Performance & Rule Compliance :

- Utilized MATLAB programing to compare tube geometry and material meet Baja SAE regulation
- Virtual design review sessions with John Hopkins and Virginia Tech for early design cycle questions
- In-person design review session with TAMU Baja SAE to ensure chassis rule compliance before CAD freeze
- Utilized social media and online networking programs to find design inspiration through other chassis and to compare various archetypes
- Torsional rigidity, roll, and impact simulations utilizing Ansys

Manufacturing:

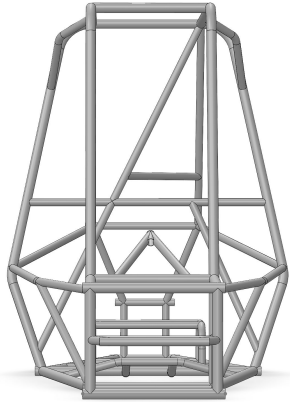
- Mobile jig + chassis manufacturing setup
- $\frac{1}{2}$ scale chassis model
- Sought to pre-order complex notches

Compliance:

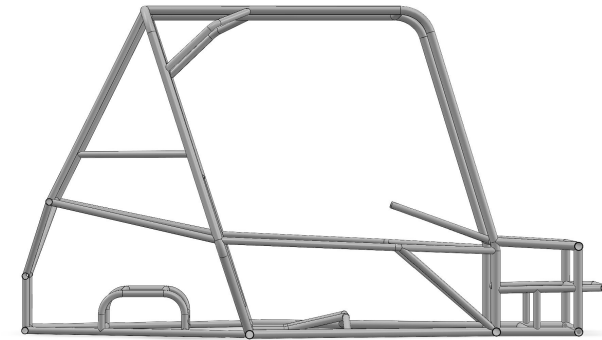
- Used two driver models fitting the top 5th and 95th percentiles to ensure driver fit within the vehicle

CHASSIS / ERGONOMICS

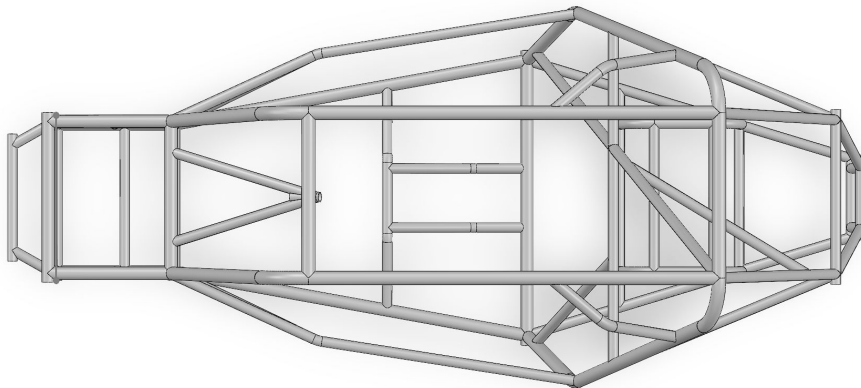
Chassis CAD – Front View



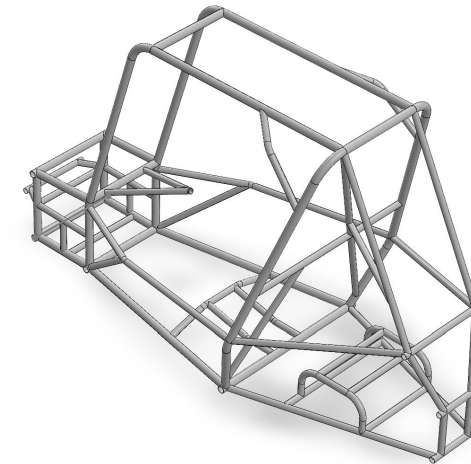
Chassis CAD – Left Side View



Chassis CAD – Top View



Chassis CAD – Isometric View



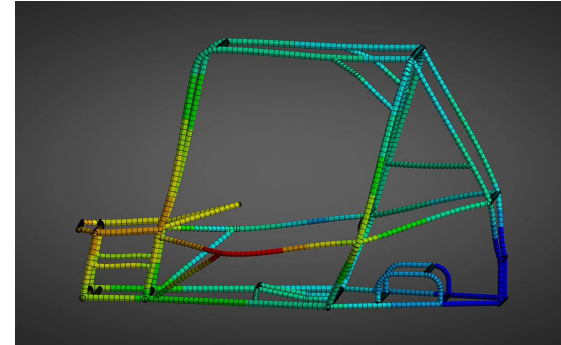
CHASSIS / ERGONOMICS

Chassis Data Collection for Analysis

Anslys Simulation Study (Static)

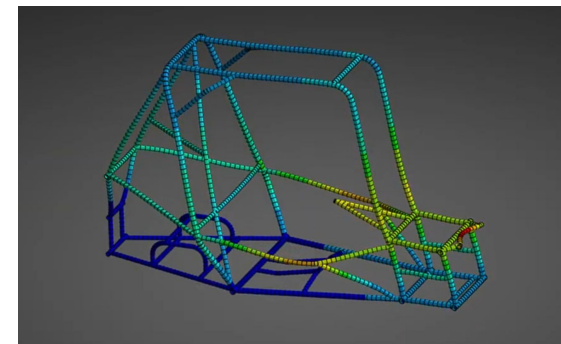
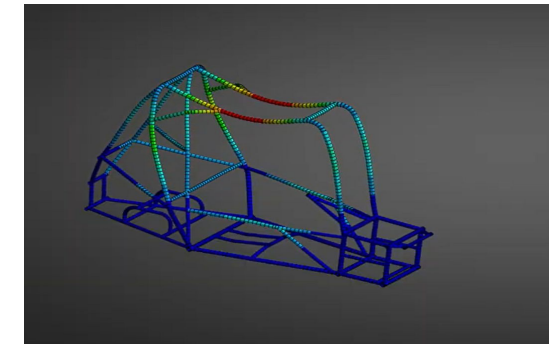
- **Frontal Collision**
 - 10000 lbf force; FoS ~ 1.3 (adequate)
- **Side Impact**
 - 5000 lbf force; FoS ~ 1.6 (adequate)
- **Rear Impact**
 - 10000 lbf force; FoS ~ 1.5 (adequate)
- **Roll Strength**
 - 5000 lbf, 1m fall; FoS ~ 4 (adequate)
- **Torsional**
 - Front and rear analysis; fixed either front or back; 5000 kN opposing forces
 - Front FoS ~ 1.2; Rear FoS ~ 1 (adequate)

Chassis Analysis Used ANSYS to simulate a front impact and validated the front bulkhead design
Used Solidworks static FEA study to validate that the engine support hoop in the rear would support the engine and powertrain.
Ran Ansys FEA to confirm how many tubes we could change from primary to secondary, especially in the rear bracing Summary



- Suspension pickup points opposite of collision
- Analyzed at member junctions and critical areas of frame (SIM).

- Floor member as fixed supports, RHO as force taking members
- 1m rollover/fall, 0 m/s at peak



- Rear suspension pickup points are fixed; front bumper as force taking member.
- Overall structure was analyzed for FoS

CHASSIS / ERGONOMICS

Chassis Testing / Validation

Beam/Material Selection

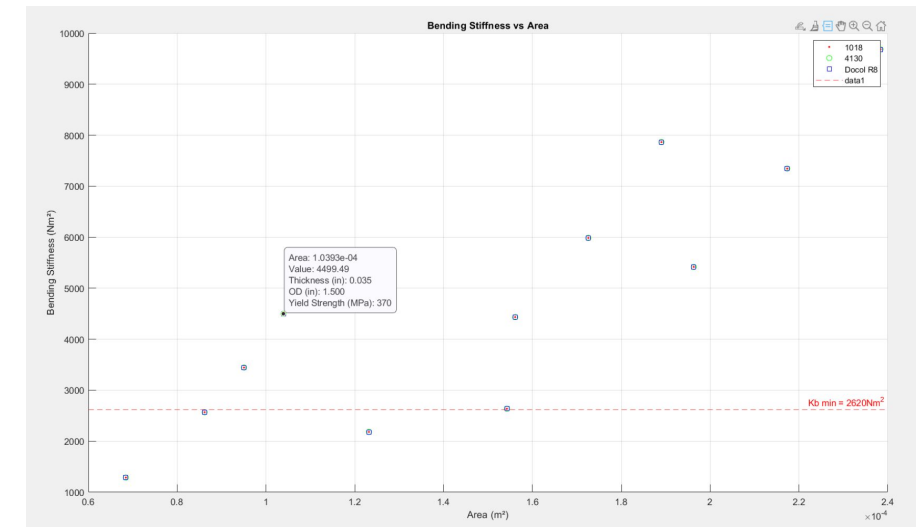
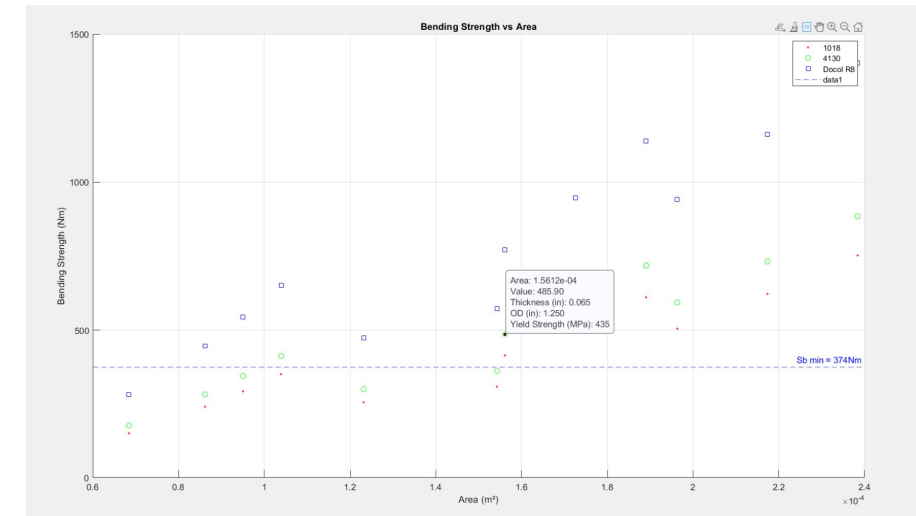
1. Swept tube outer diameter (1.0"–2.5") and wall thickness (0.049"–0.134") across AISI 1018, AISI 4130, and Docol R8 steel. For each combination, computed bending stiffness and bending strength.
2. Filtered out results that did not meet the requirements of minimum bending stiffness ($K_b \geq 2620 \text{ N}\cdot\text{m}^2$) and minimum bending strength ($S_b \geq 374 \text{ N}\cdot\text{m}$)

Our bending strength given by the chosen chassis materials yields about $435 \text{ N}\cdot\text{m}$, which is greater than the required value of $374 \text{ N}\cdot\text{m}$.

Our bending stiffness given by the chosen chassis materials yields about $4499.49 \text{ N}\cdot\text{m}^2$, which is greater than the required value of $2620 \text{ N}\cdot\text{m}^2$.

Other considerations included manufacturability, cost, weight, etc...

Chassis Correlation of Analysis and Testing



CHASSIS / ERGONOMICS

Chassis Manufacturability

Sectioned Assembly

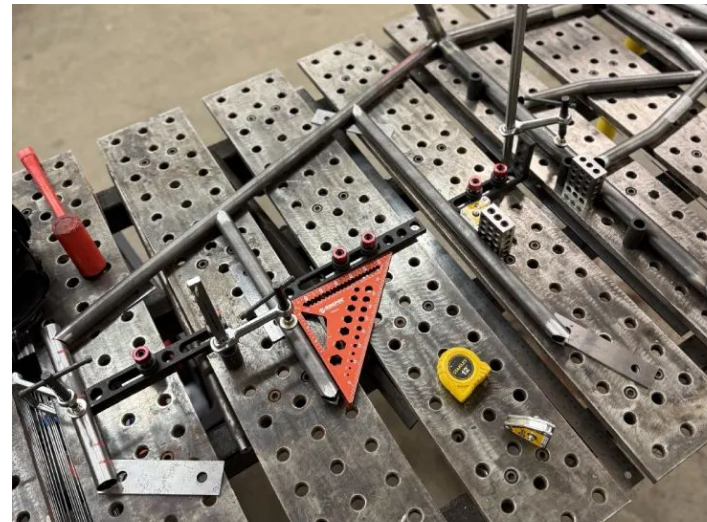
- Divided chassis into smaller subassemblies which improved weld accessibility and lowered complexity

Modular Welding Jig

- $\frac{5}{8}$ " Stronghand fixture stops and clamps constrained chassis to welding table
- Triangle fixtures, magnets, and angle finders accurately located tubes and joints during assembly
- Fixture points derived from CAD geometry

Tubes

- Bent tubes outsourced from VR3 Engineering
- Straight tubes cut and notched in-house with JD2 tube notcher
- Tubes tig welded



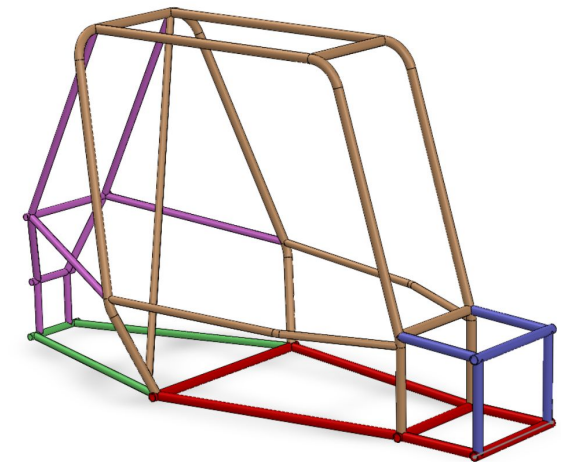
Chassis Serviceability

Accessibility

- Engine and CVT removable without disassembling surrounding chassis components

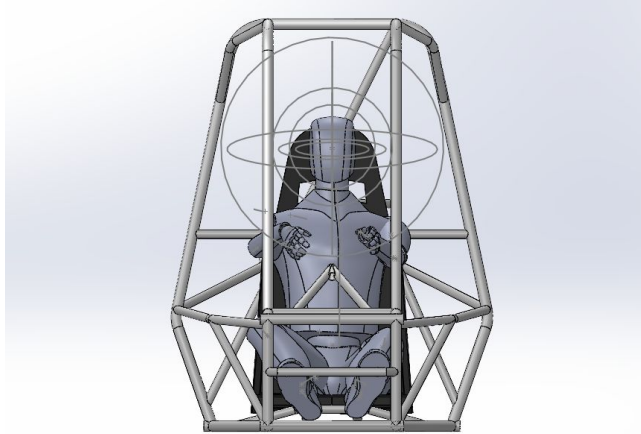
Fasteners

- Implemented consistent quarter-turn fastener size and type for quick removal of body panels without sacrificing security to the frame

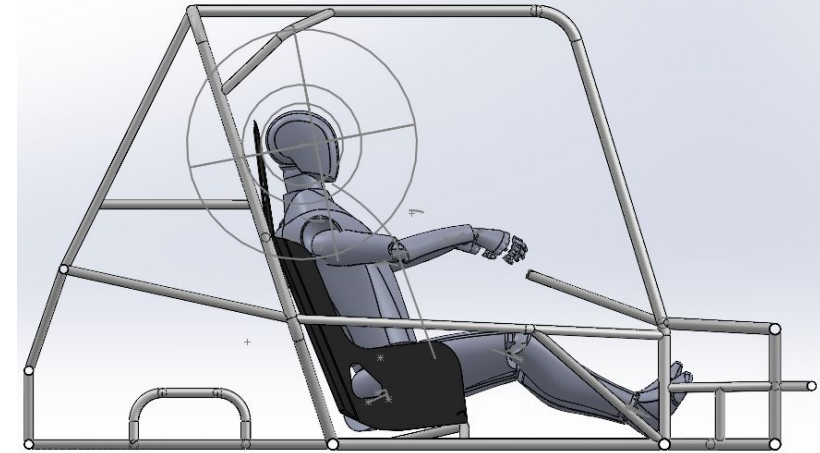


CHASSIS / ERGONOMICS

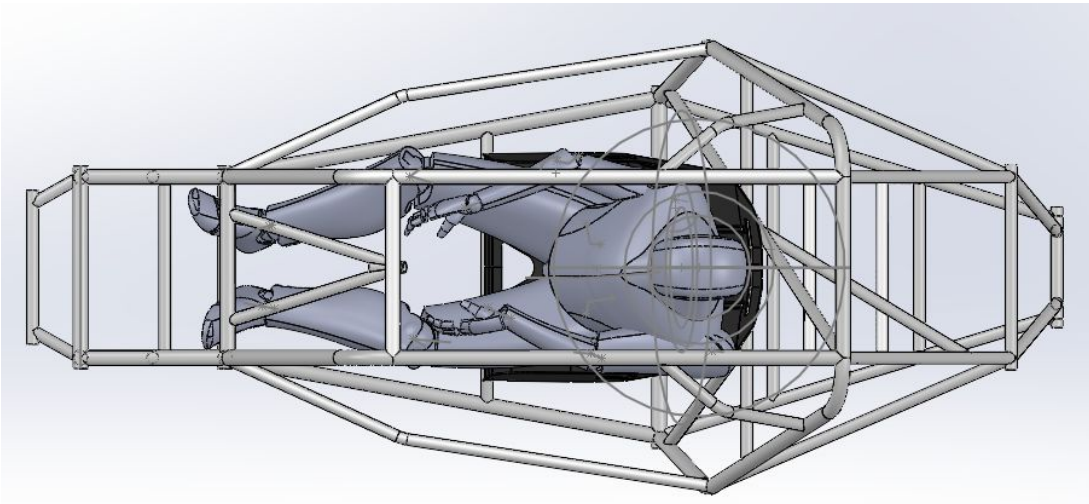
Ergonomics CAD – Front View



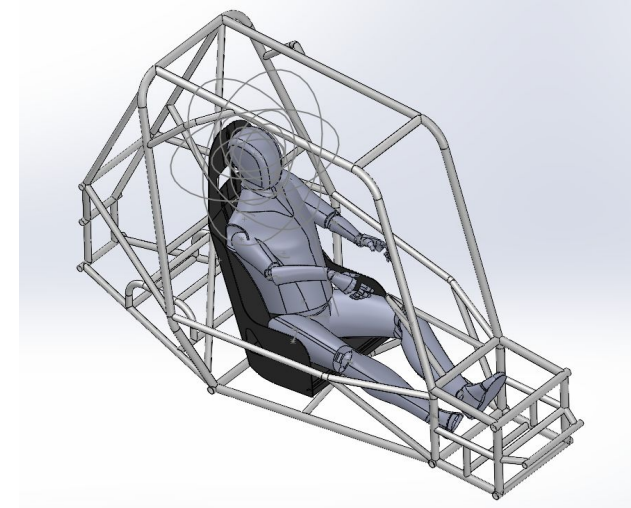
Ergonomics CAD – Left Side View



Ergonomics CAD – Top View



Ergonomics CAD – Isometric View



CHASSIS / ERGONOMICS

Ergonomics Data Collection for Analysis

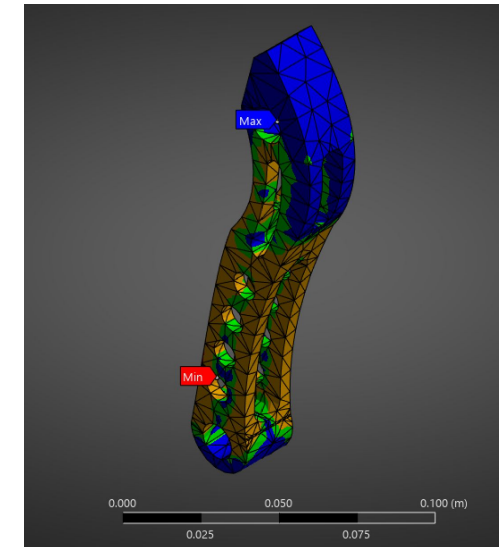
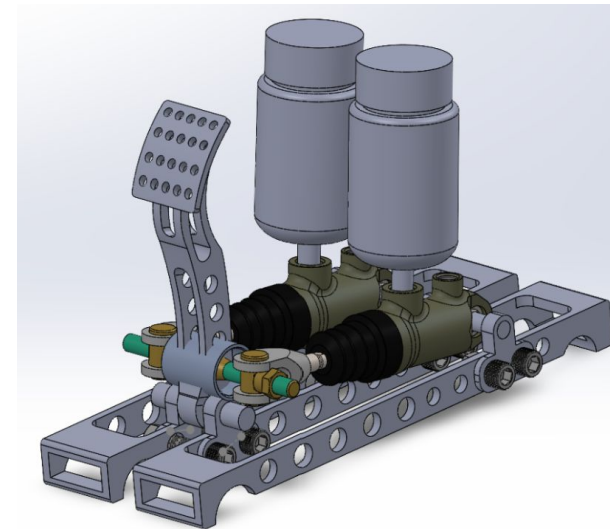
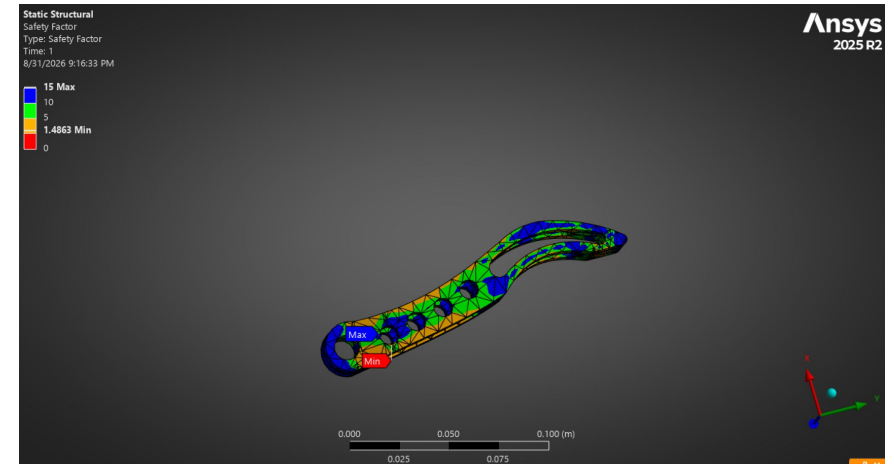
Throttle & Brake System - Pedal Box Design

1. Mounting bracket creates adjustable settings for driver personalization by feel and bodily proportions
1. Aluminum 6061 for its lightweight, durability, and low cost
2. Brake system has a pedal ratio of 6
 - a. 500 N at the edge of the brake pedal
 - b. 100 N acting on the pedal via master cylinders
3. Throttle System
 - a. 500 N inputted on the edge of the brake pedal
 - b. 80 N at the intake string onto the throttle pedal

Seat Design

1. Measured anthropometric data of each driver to determine optimal points of common joint intersections
2. Created custom driver models to simulate driver fit and packaging before fabrication
3. Aluminum 6061 chosen due to its lightweight, low cost, and ease of manufacturability

Ergonomics Analysis Summary



CHASSIS / ERGONOMICS

Ergonomics Manufacturability

Off-the-Shelf Parts

- Kirkey Seat, Azusa Steering wheel, and Racing Pedal box were purchased off-the-shelf
- Experienced machinists and other Baja teams suggested against custom fabricated parts for a first-year team
- Before this change, we made several designs for each part, including brake & throttle system

Custom Fabricated Designs

- While we unfortunately had to buy the brakes & throttle system off the shelf, it is originally meant to be manufactured
- couldn't be manufactured due to timeline constraints
- Also over complication in system due to pneumatics in throttle, and brake pedal requiring 5 axis CNC to manufacture
- Next design cycle will involve lightening, reducing contours on the pedal arm & replace pneumatics with coil springs

Ergonomics Serviceability

Ease of Access

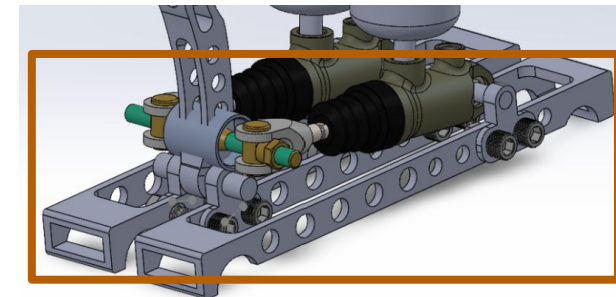
- Quarter-turn fasteners on each body panel for ease of access into tight spaces such as the front bulkhead
- Fire extinguisher mounted for quick access by the driver and on-track team

Driver Adjustability

- Pedal box allow drivers of varying leg lengths to select their preferred angle and length settings

Subsystem Integration

- Coordinated with other subteams to ensure clearance between assemblies
- Packaged assemblies to allow other subteams to access their components with ease
- Brake light pressure sensor mounted for both electronics and ergonomics troubleshooting



CHASSIS / ERGONOMICS - Appendix

Half-Scale Manufacturing Prototype

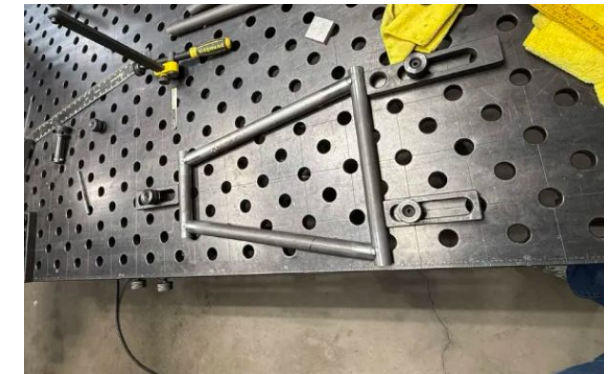
Purpose

- Validate the planned chassis manufacturing and assembly process before full-scale fabrication

Validation and Learnings

- **Fixturing:** Practiced using stops and clamps to maintain chassis geometry
- **Welding:** Confirmed TIG welding method and observed material response
- **Tube Preparation:** Practiced cutting, notching, and coping tubes for joint fitment
- **Fitment, Tolerances, and Angles:** Evaluated manufacturing tolerances and practiced jiggging accurate angles
- **Assembly Sequence:** Identified efficient order for fixturing, tacking, and final welding

Images



2026 Baja SAE Design Review Briefing (DRB)

School Name

University of Texas at Austin

2026 Car Number(s)

185

ELECTRONICS

Requirements

Original requirements, set February 2026

- Log vehicle sensor data to onboard microSD on every test run
- Condition every sensor input to the 3.3 V limit of the RP2040
- Allow any sensor to be replaced at the track without reworking the board
- Keep the board hand assemblable by a first year team with no reflow oven

Requirements as built, after the July budget reduction

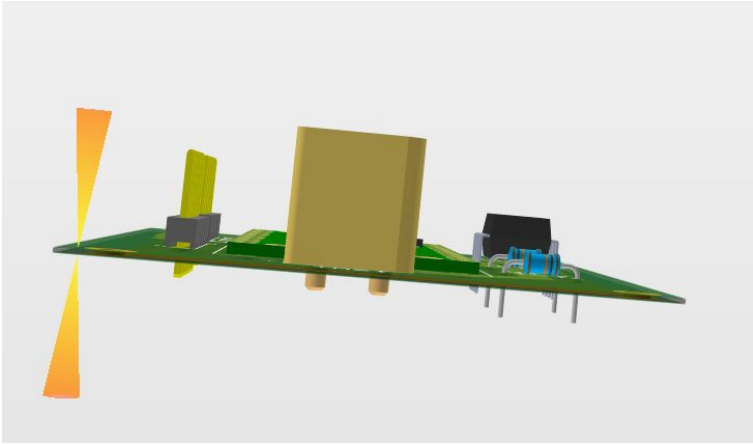
- Log six axis inertial data to onboard microSD on every test run
- Single 12 V vehicle supply on a tool free connector
- Front and rear engine kill switches on a circuit isolated from the board
- Keep every removed channel designed, routed and documented so it can be populated when budget allows

Research and Development

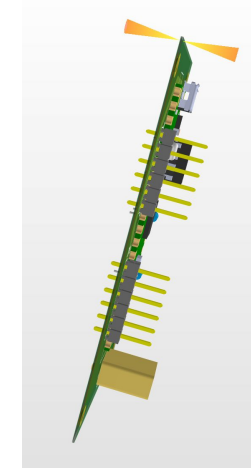
- This is a first year team with no historical data. Every subsystem this year was sized from calculators and competitor benchmarking rather than measured loads. The ECU exists so next year starts from measurement instead.
- Sensor selection was driven by the 3.3 V logic limit and by the fact that every sensor lives somewhere else on the car, so all of them connect through headers rather than mounting to the board.
- The schematic was split into nine subsystem sheets so five people could work in parallel. Each sheet has one owner and the sheet boundary is the interface.
- A full seven channel ECU was designed, routed and released to fabrication on 29 May. A budget reduction in July cut the build to a reduced board. See the Design Scope slide for what was removed and what survives on file.
- Design files are version controlled in git. 69 commits between February and July 2026, which is where the revision history on these slides comes from.

ELECTRONICS

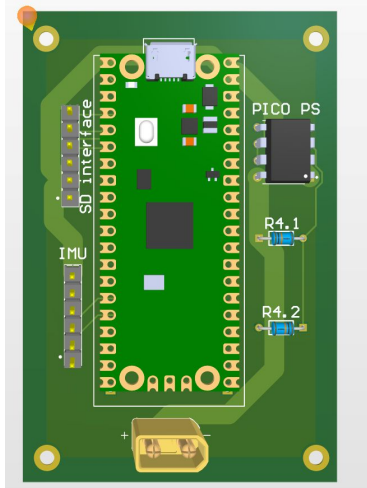
Sensor ECU CAD - Front View



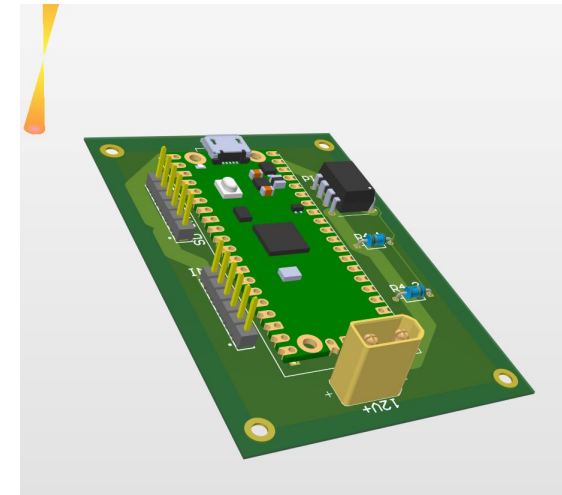
Sensor ECU CAD - Left Side View



Sensor ECU CAD - Top View



Sensor ECU CAD - Isometric View



ELECTRONICS

Sensor ECU Analysis Summary

As built board, nine components on two layers

- Raspberry Pi Pico (RP2040) as the controller, generating its own 3.3 V rail
- Six axis IMU on I2C, connected through a 1 x 6 header at 2.54 mm pitch
- microSD interface on SPI, using MOSI, MISO, clock and chip select
- Single 12 V vehicle supply on an XT30 connector rated 15 A and 500 V
- 5 V rail derived from the 12 V input by a 35k and 25k divider buffered by a TLV2372 rail to rail op amp

Signal conditioning principle

- The Pico accepts 3.3 V maximum. Every derived rail and sense line on this design is divided to land at or below that limit.
 - 12 V through 35k and 25k gives 5.00 V for the sensor rail
- The same method sized the channels that were designed but not populated: wheel speed at 5 V through 5.1k and 10k gives 3.31 V, brake at 12 V through 10k and 3.9k gives 3.37 V.

Sensor ECU Data Collection for Analysis

Channels logged on the car

- Six axis IMU. Body accelerations and angular rates, logged to microSD.
- This feeds chassis load analysis and ride behaviour, and gives suspension a measured input to compare against the calculator predictions used to size the links this year.

Channels designed but not populated

- Two Hall effect wheel speed, brake pressure, CVT temperature. All four are drawn, conditioned and routed on file. None are on the car.

Known gaps on the as built board

- No fuse on the 12 V input and no bulk decoupling capacitor. Both were on the full design and came off with the reduction. Both are first items for the next revision.

Logging targets 100 Hz, the BNO055 fusion output ceiling, writing 32 byte packed binary records at 3.2 kB/s for about 11.5 MB per hour, or 46 MB over a four hour endurance run, which is 0.01% of the card's 30 MB/s rated write speed

ELECTRONICS

Sensor ECU Correlation of Analysis and Testing

Analysis predicts the following. Measured values to be completed at bench test.

5 V rail (no load): Predicted 5.00 V (from the 12 V rail through a 25 kΩ / 60 kΩ divider)

5 V rail (SD card writing): Predicted to sag (buffered by an op-amp output stage instead of a dedicated regulator)

3.3 V rail (at Pico): Predicted 3.30 V (from the Pico's internal regulator)

Linear acceleration (board level): Predicted 0.0 m/s² (BNO055 sensor fusion strips out gravity)

Quaternion (board level): Predicted (1, 0, 0, 0) (identity orientation)

Sample rate: Predicted 100 Hz (the BNO055's maximum fusion output rate)

File growth: Predicted ~11.5 MB/hr (32-byte records logged at 100 Hz)

The 5 V rail is our highest-risk prediction. Since it's driven by a TLV2372 op amp buffering a voltage divider rather than a true regulator, its output stage will likely struggle during SD card write bursts that exceed 100 mA. We hope to add a linear regulator instead for the first bench test

Sensor ECU Testing / Validation

Design decisions made for the vibration and electrical noise environment

- Sensors connect through headers, so a sensor damaged on course is a plug swap rather than a rework job
- XT30 supply connector is rated 15 A and 500 V, well beyond the load, and is a positive latching fit
- Two layer board with no fine pitch parts, so every joint can be inspected by eye

Suggested bench validation, all achievable in an afternoon

- Sustained log run on the bench, then confirm the file count and sample continuity
- Rail voltage under SD write burst, on a scope rather than a multimeter
- Shake or tap test while logging, to confirm the connectors stay seated and samples keep landing
- The board is fabricated and bench validation is scheduled for 9/3/2026. No vehicle data has been logged yet

ELECTRONICS

Sensor ECU Serviceability

- Every sensor connects on a header instead of being soldered to the board, so a failed sensor is a plug swap in the pits rather than a rework job with a soldering iron.
 - IMU on a 1 x 6 header, 2.54 mm pitch
 - microSD on its own interface header
 - Vehicle supply on an XT30 connector, tool free
- Sensors were originally drawn as bare ICs mounted to the board. On 8 March every one of them was moved to a connector. The sensors live elsewhere on the car, so mounting them to the ECU was never going to work.
- The four removed channels all use the same header approach, so populating them later needs no layout change to the sensor side.

Sensor ECU Manufacturability

- Through hole packages were chosen over fine pitch wherever there was a choice, because the board is assembled by hand. The op amp was moved from a VSSOP package to PDIP for exactly this reason.
- Polygon connections use thermal relief. A pad tied straight into a copper pour sinks enough heat during hand soldering that the joint may never properly wet.
- Two layer board. Cheaper and faster to fabricate than four layer, and viable here because there is one analogue concern, divider accuracy, and no high speed digital buses.
- Every BOM item was checked against live supplier data and confirmed at volume production status, so nothing specified was heading for obsolescence.
- Gerbers and drill files are generated from source rather than committed, so fabrication outputs cannot drift out of step with the design.
- 3D STEP models for the Pico and the XT30 connector are in the project, so connector clearance was checked against the board before fabrication rather than after.

ELECTRONICS

Power Distribution Analysis Summary

As built

- Single 12 V vehicle supply into the board on an XT30 connector, rated 15 A and 500 V
- 5 V sensor rail derived on board from the 12 V input, feeding the IMU and the SD interface
- 3.3 V logic rail generated by the Pico itself
- Front and rear engine kill switches on their own circuit, isolated from the board, so they function regardless of the state of the electronics

Net naming and rail distribution

- On 1 March every rail was converted from an ordinary port to a global power port across all nine sheets. An ordinary port only connects through an explicit sheet entry, so carrying rails that way means hand wiring every rail through every sheet boundary. Converting them removed that whole class of silent wiring error.

Designed but not populated

- The full design carried two separately fused 12 V batteries, one for sensors and one for the brake circuit, so an electrical fault on the sensor side could not affect braking. Both batteries, both fuses and the bulk decoupling came off with the budget reduction.

Power Distribution Data Collection for Analysis

Predicted loads, calculated. Measured values pending bench test.

- 5 V rail, idle: approximately 47 mA. Pico around 30 mA, BNO055 around 12 mA, SD card idle around 5 mA
- 5 V rail, SD card writing: up to approximately 142 mA
- 12 V input current: approximately the same as the 5 V load, since the rail is generated by a linear buffer rather than a switching regulator
- Resistor divider current: 200 μ A, from 12 V across 60 k Ω
- Op amp dissipation: 0.33 W at idle, rising to approximately 0.99 W during SD writes, from 7 V dropped across the output stage

Next revision

- Fuse on the 12 V input, sized against measured worst case current
- Bulk and high frequency decoupling restored at the supply entry
- Confirm the 5 V rail can hold up under SD write current, or replace the buffer with an LDO

ELECTRONICS

Power Distribution Correlation of Analysis and Testing

Analysis predicts the following. Measured column to be completed at bench test.

- 12 V input current, idle: predicted 48 mA
- 12 V input current, SD writing: predicted 142 mA
- Op amp case temperature after sustained logging: predicted hot, 0.99 W in a PDIP-8
- 5 V rail during an SD write burst: predicted to sag, op amp bandwidth is 3 MHz and the burst is faster
- Harness voltage drop: to be measured, depends on final wire gauge and routing length

The op amp is the component we expect to fail these predictions. It buffers a resistor divider rather than regulating, so it carries the full 7 V drop as heat and cannot respond to fast transients

Power Distribution Testing / Validation

Planned Validation:

- Kill switch function check, front and rear, engine stops in both cases
- Sustained logging run with a thermocouple or thermal camera on the op amp and the 12 V trace
- Supply brownout check, drop the input voltage and find where logging stops, then compare that to worst case cranking voltage
- Connector retention check, tug test on the XT30 under vibration

Bench validation is scheduled for 9/3/2026. The kill switch check is the priority, since its crucial for safety.

ELECTRONICS

Power Distribution Serviceability

- The 12 V supply connects on an XT30 and can be disconnected without tools, so the board can be isolated from the vehicle in seconds.
- Kill switches are on their own circuit, so they keep working regardless of the state of the sensor electronics.
- The board mounts as a unit and can be removed from the car without disturbing the harness, since every connection is a connector rather than a solder joint.
- The battery is mounted to the rear of the firewall next to a tab close to the chassis to allow for easy wire routing along the chassis.
- The ECU is mounted nearby as well for the convenience of a relatively centralized electronics system.
- One kill switch is mounted at the top of the chassis as specified in the rulebook while the other is mounted on the dash for ease of access to our driver.
- Removal of the board will require unscrewing components of the pcb while removing the battery will require cutting zipties that will be used to mount it in place.
- All components are coated in a layer of epoxy for environmental protection.

Power Distribution Manufacturability

- XT30 is a through hole part, widely stocked, inexpensive, and solderable by hand.
- Kill switches are standard catalogue parts with no custom tooling.
- 3D STEP models for the Pico and the XT30 are in the project, so mating clearance was verified in CAD before the board was ordered.
- Harness connections are crimped with heat shrink at every joint, using stranded wire throughout so vibration can't crack any conductors.

ELECTRONICS

Design Scope and Budget Reduction

What happened

- A full seven channel sensor ECU was designed, routed and released to fabrication on 29 May 2026. Thirty one components on a two layer board.
- In July the electrical budget was reduced. Rather than cancel the system, the board was cut back to the smallest configuration that still produces useful vehicle data, and built.

What the car carries

- Nine components. Pico, six axis IMU, microSD logging, single 12 V supply, and the isolated kill switch circuit.

What came off

- Two Hall effect wheel speed channels
- Brake pressure sensing
- CVT temperature sensing
- Driver LCD display
- Cooling fan driver
- Second battery, both fuses, and the bulk decoupling

What Was Designed, Validated and Retained

- Nothing that was cut was thrown away. Every removed channel is drawn, conditioned, routed and held in version control, so populating it is a build decision rather than a design task.

Retained on file and ready to populate

- Signal conditioning sized for each channel. Wheel speed at 5 V through 5.1k and 10k gives 3.31 V. Brake at 12 V through 10k and 3.9k gives 3.37 V.
- Current based trace width rules. Ground and 12 V battery at 3 mm preferred with 5 mm maximum on the battery net, fan gate drive at 0.75 mm, all signal nets at 0.254 mm.
- The complete four revision cooling fan driver, described on the next slide.
- Custom footprints and symbol libraries for the LCD, the SD socket and the Pico.

Why this matters

- Next year's team inherits a validated seven channel design and a build decision that is ready.

ELECTRONICS

Sensor ECU Design Progression

- 21 Feb One flat schematic, 37 components on a single sheet
- 21 Feb Split into nine subsystem sheets for parallel work
- 24 Feb Sensor dividers rescaled to land 5 V inputs at 3.31 V
- 01 Mar All rails converted to global power ports across eight sheets
- 06 Mar Fan driver reaches its third of four revisions
- 07 Mar Design rules written, trace widths sized by current
- 08 Mar Sensors moved off board onto connectors
- 08 Mar Full design complete, 31 components on two layers
- 29 May Gerbers and drill files released to fabrication
- 14 Jul Budget reduction, board rebuilt at nine components
- 28 Jul As built board finalised

All of it is in version control. 69 commits, five contributors.

Worked Example: Cooling Fan Driver

Designed and validated, not populated on the as built board. Included because it is the clearest record of how the team iterated.

- The fan was the largest load on the full design at 12 V and 2.5 A. It took four revisions.
 - Rev 1: MOSFET switched directly from a microcontroller pin
 - Rev 2: added a dedicated gate driver and a 22 ohm series gate resistor
 - Rev 3: added local decoupling at the gate driver
 - Rev 4: replaced the driver IC with a small signal MOSFET level shifter driving a 100 V power MOSFET
- Rev 1 was wrong because 3.3 V cannot fully turn on a power MOSFET that wants 10 V at the gate. Half on, it sits in its linear region and turns power into heat instead of switching.
- Rev 4 traded the driver IC for two commodity through hole parts doing the same job. It switches more slowly, which does not matter for a cooling fan on low frequency PWM, and it is cheaper, easier to hand solder and easier to replace at competition. For a first year team building boards by hand, that was the right trade.

2026 Baja SAE Design Review Briefing (DRB)

School Name

University of Texas at Austin

2026 Car Number(s)

185

DRIVETRAIN

Requirements

Requirements

- Rear Wheel Drive
- Transverse Engine
- Transmits power from engine to wheels

Goals

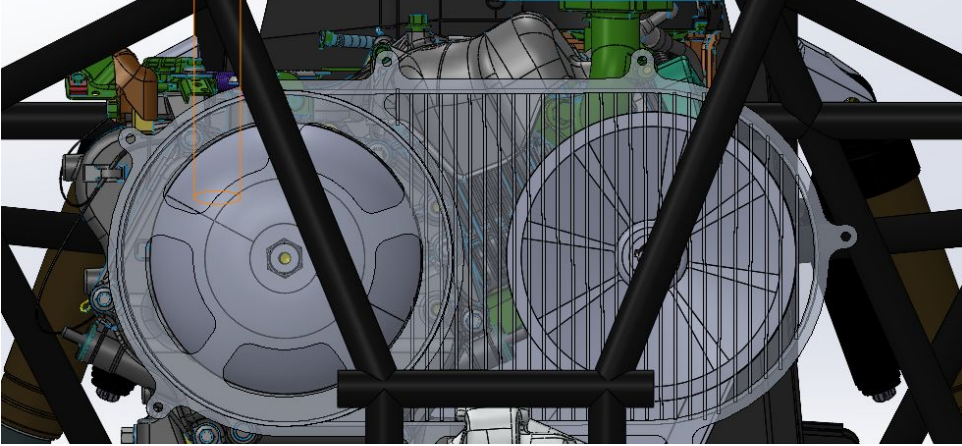
- Pass technical inspection
- Serviceable CVT & Final Drive

Research and Development

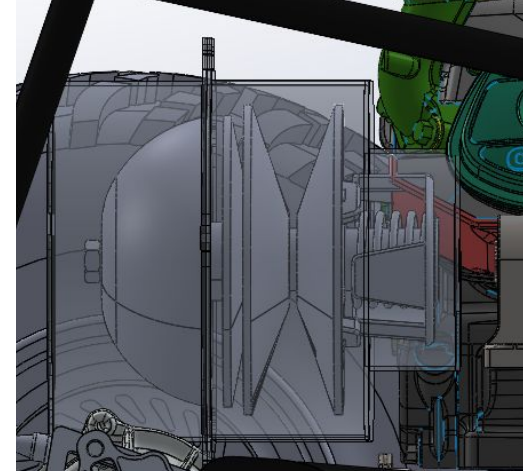
- Decision matrices for each assembly
- Gear & shaft manual calculation
- FEA & CFD analysis of all guarding
- Cross-checking differential calculations with original OEM application.

DRIVETRAIN

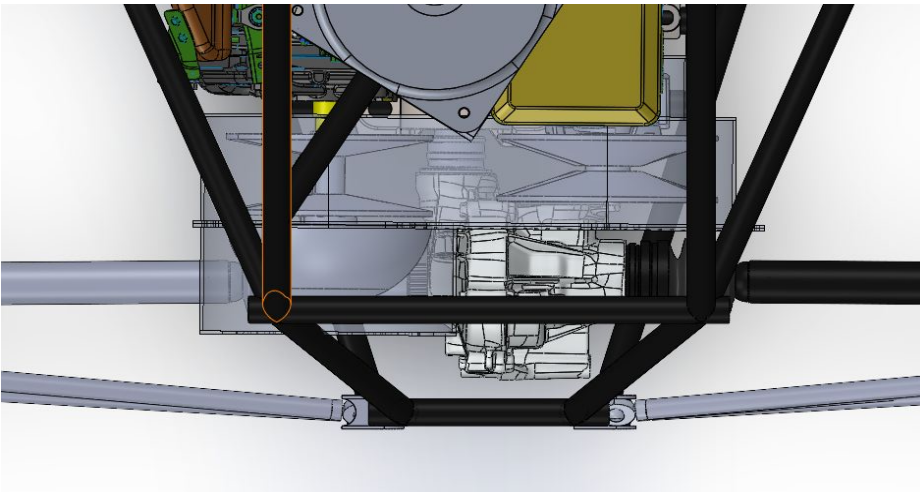
CVT / Transmission CAD – Front View



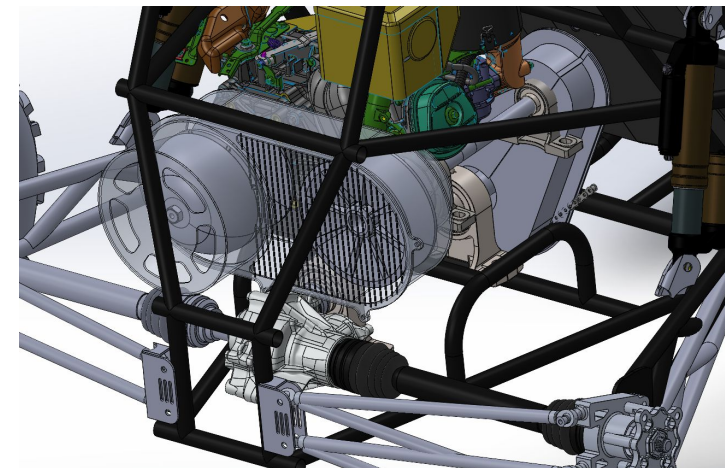
CVT / Transmission CAD – Left Side View



CVT / Transmission CAD – Top View



CVT / Transmission CAD – Isometric View



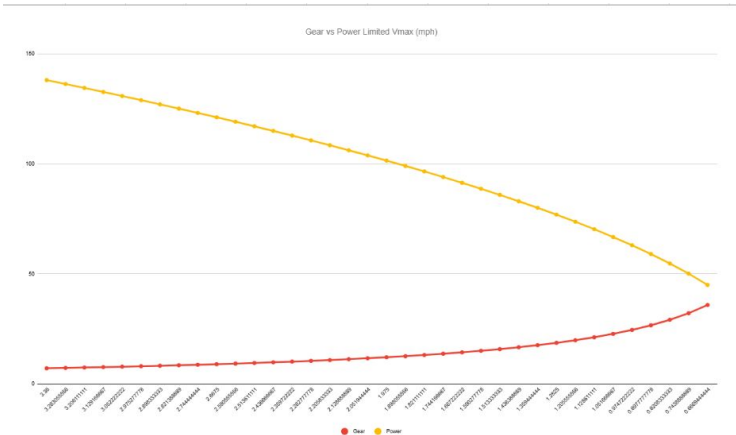
DRIVETRAIN

CVT / Transmission Testing / Validation

Method: CVT Tuning Calculator (excel)
Results:

ENGINE-CVT TABLE			
	CVTech	CVTech	Example cvt
HI	0.43	1	
LOW	3	1	
Efficient ratio	0.89777		Chosen from chart below
Set Engage RPM	1800		Set yourself
Engine Max RPM	3650		Dont blow it up
Engine Peak Torque (lb-ft)	18.5		2400
Engine Top End Torque (lb-ft)	13.75		
	HIGH RATIO	LOW RATIO	EFFICIENT
Ratio	0.43	3	0.89777
Total Drivetrain Reduction	5.2653	36.7347	10.9931
Secondary Pulley RPM	8488.37	1216.67	4065.63
Tire RPM	693.2170543	99.36111111	332.0263913

	Driver (teeth)	Driven (teeth)	Driver (teeth)	Driven (teeth)
Tooth #	21	60	21	90
Reduction	2.857		4.286	
	First Stage Reduction		Second Stage Reduction	
	12.245			
	Total Gearbox Reduction			



CVT / Transmission Correlation of Analysis and Testing

- Mapped torque multiplication from engine crank to CV axle across the full CVT range.
- Determined torque and speed at each drivetrain stage under full load.
- Optimized chain reduction to balance top speed, axle torque, and component margins.
- Verified axle torque stays within the differential's OEM operating range.

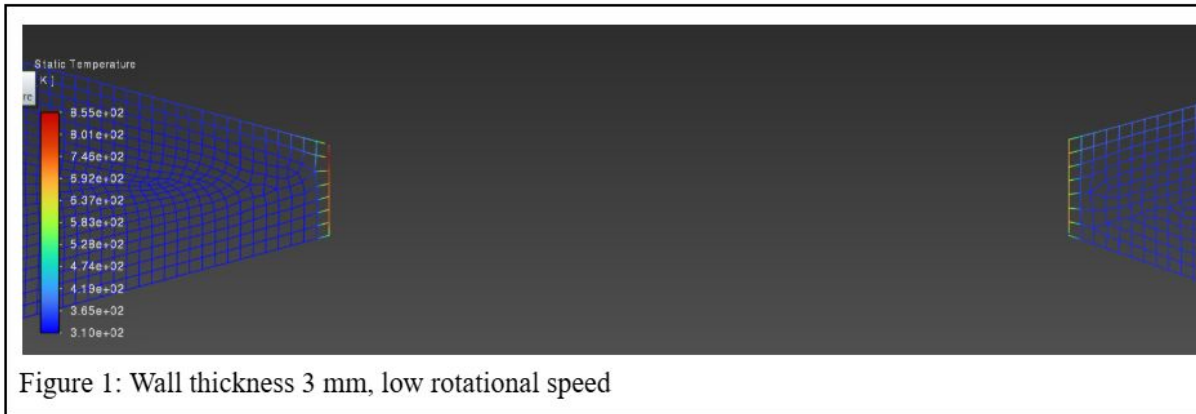
DRIVETRAIN

CVT / Transmission Collection for Analysis

CFD Analysis:

- Fan necessity determined by max temperature
- Data collected through streamline scenes

Results: Detailed methodology and engineering rationale documented for design validation and future iteration can be accessed [here](#).



CVT / Transmission Analysis Summary

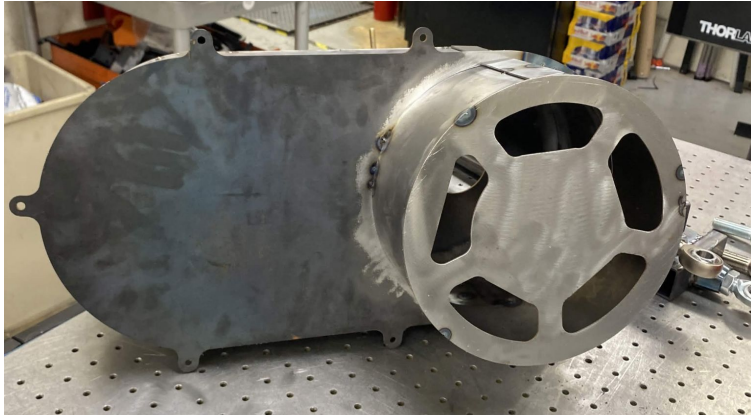
CFD Analysis Summary

- CVT meets performance requirements for short-duration operation.
- Worst-case model reaches 360 K critical temperature in ~250 s
- Extended maximum-load operation presents a significant thermal risk.
- Current airflow direction limits cooling performance.
- Recommend casing geometry changes to direct airflow through internal CVT channels.
- Improved convection will increase allowable maximum-load operating time.

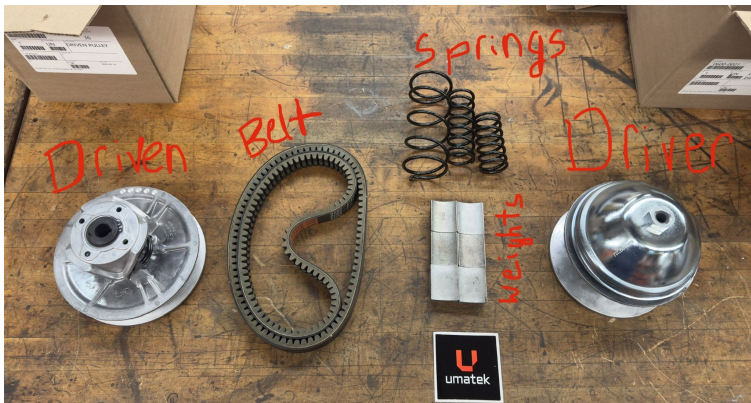
DRIVETRAIN

CVT / Transmission Manufacturability

Serviceable CVT Casing Design

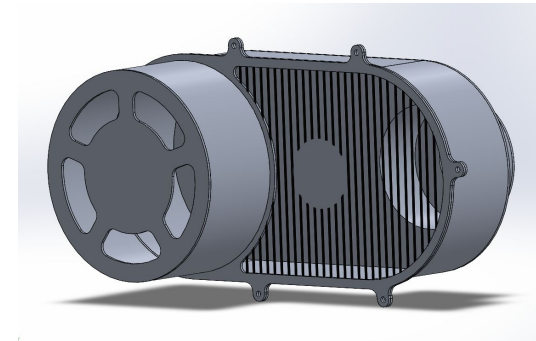


CVTech CVT



CVT / Transmission Serviceability

- We only expect to service or replace the belt.
 - **Belt:** Accessible via screws on face of CVT casing
 - *We do not aim to replace it during endurance*
- **CVT Primary and Secondary**
 - **Mounting:** Mounted with single bolt and washer, torqued to manufacturer recommended spec
 - *Estimated 2 hour replacement time if needed*



DRIVETRAIN

Front Drive, Prop Shaft, Front Hubs and Axles Data Collection for Analysis

Vehicle loading:

- Determined vehicle mass, weight distribution, and front/rear axle loads under acceleration, braking, and obstacle loading.

Powertrain loads:

- Max drivetrain torque through the prop shaft and axles using engine torque and CVT/gear reduction ratios.

Material properties:

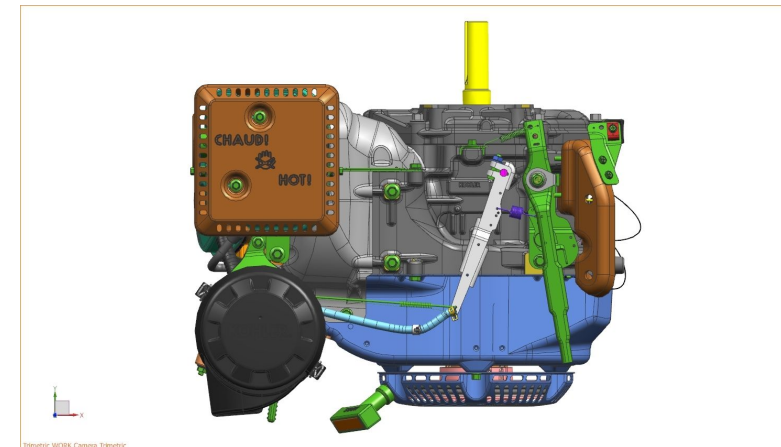
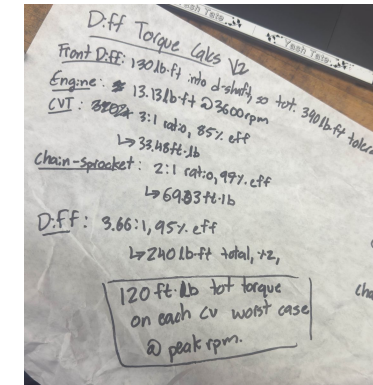
- Defined material properties and applied yield strength, elastic modulus, and Poisson's ratio for FEA.

Load cases:

- Established worst-case torque, impact, and wheel loads for acceleration, rock crawl, hill climb, and uneven-terrain conditions.

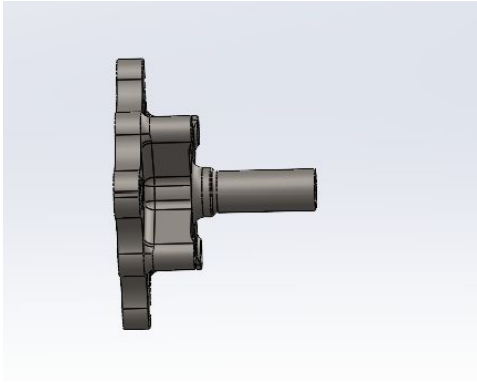
Front Drive, Prop Shaft, Front Hubs and Axles Analysis Summary

Shaft C to C (mm)	Belt #	Pitch length (mm)	Min ratio	Max ratio	Ratio range
200	BD52-2167-S	835	3.0 : 1	0.43 : 1	6.97
232	BD52-2179-S	895	3.0 : 1	0.43 : 1	6.97
242	BD52-2183-S	915	3.0 : 1	0.43 : 1	6.97
299	KE52-2205-S	1025	3.0 : 1	0.43 : 1	6.97



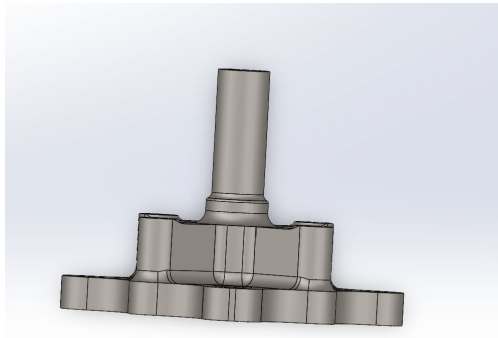
DRIVETRAIN

Front Drive, Prop Shaft, Front Hubs and Axles CAD – Front View



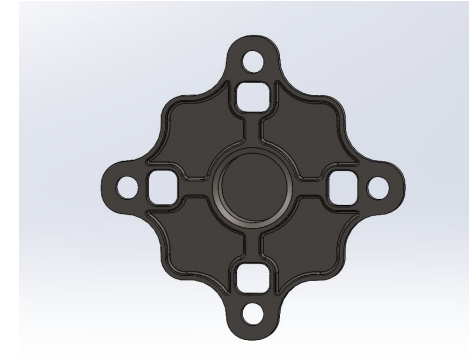
Front Hubs, Front Axles, N/A - Front Drive, N/A - Prop Shaft

Front Drive, Prop Shaft, Front Hubs and Axles CAD – Top View



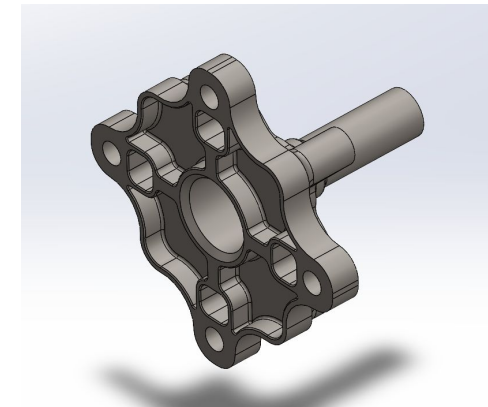
Front Hubs, Front Axles, N/A - Front Drive, N/A - Prop Shaft

Front Drive, Prop Shaft, Front Hubs and Axles CAD – Left Side View



Front Hubs, Front Axles, N/A - Front Drive, N/A - Prop Shaft

Front Drive, Prop Shaft, Front Hubs and Axles CAD – Isometric View



Front Hubs, Front Axles, N/A - Front Drive, N/A - Prop Shaft

DRIVETRAIN

Front Drive, Prop Shaft, Front Hubs and Axles Testing / Validation

FEA inputs:

- Applied calculated loads and boundary conditions, with refined meshing at key stress-concentration regions.

Results:

- FEA results confirmed the drivetrain components withstand 4–5G loading with acceptable safety factors.

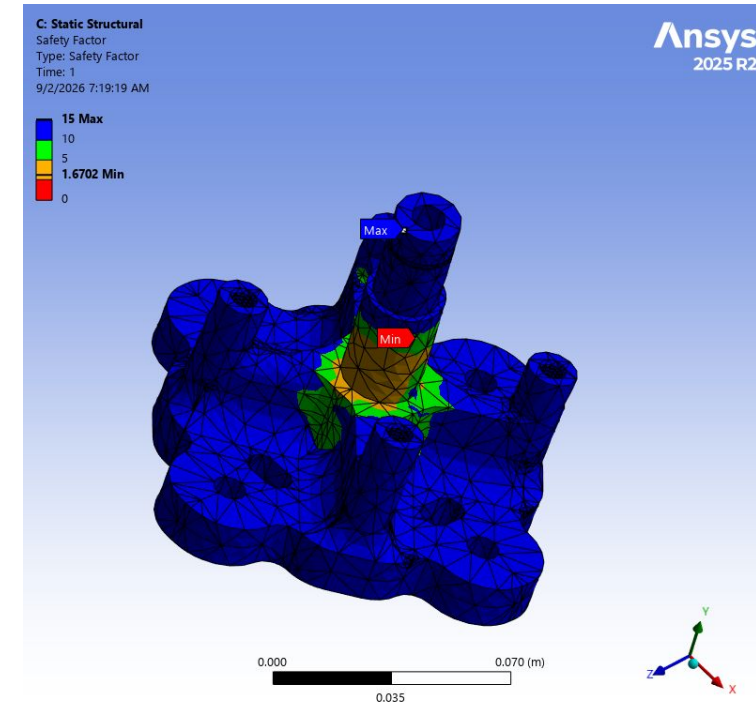
Validation:

- Validated FEA results against operating loads to confirm component strength and required safety factors.

Front Drive, Prop Shaft, Front Hubs and Axles Correlation of Analysis and Testing

Sim 5: Jump + Steering (Worst Case)

Forces	Amount	Units	Direction
Jump Vertical Load	2668.93	Newtons	+Y
Longitudinal Friction (equal split)	1509.775	Newtons	-X
Lateral Friction (equal split)	1509.775	Newtons	+Z / lateral
Hub Torque	441.005	N·m	About hub axis



DRIVETRAIN

Front Drive, Prop Shaft, Front Hubs and Axles Manufacturability

Design Methodology

- Designed for simple, repeatable machining
- Optimized geometry to reduce material removal
 - Added appropriate fillets, tolerances, and accessible features.
- Used standard bearings, fasteners, and interfaces to simplify assembly and sourcing.

Front Drive, Prop Shaft, Front Hubs and Axles Serviceability



Front Hubs

DRIVETRAIN

Reduction Box, Rear Hubs and Axles Data Collection for Analysis

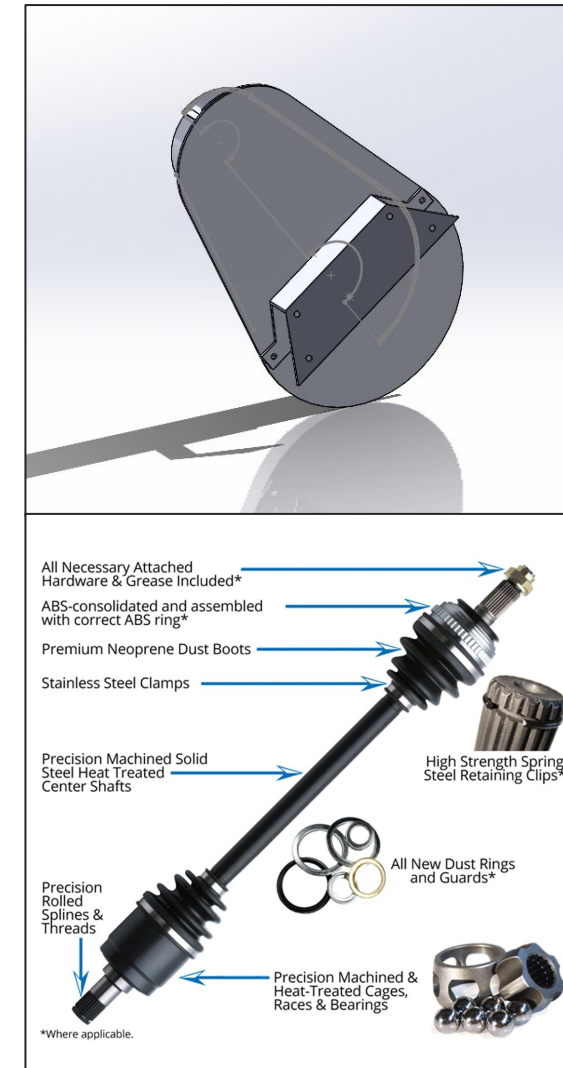
Reduction Chain Drive + Differential

- Yamaha Grizzly 660 differential ring & pinion ratio (3.66:1) confirmed via direct inspection/teardown
- CV axle shaft diameter and spline specifications pulled from OEM documentation
- Restricted-engine torque curve used as the drivetrain input
 - (Chain/sprocket final-stage ratio and estimated mechanical efficiency)

Rear Axles

- Rear axles extension plan with design review
- Analysis using FEA and design reviews

Reduction Box, Rear Hubs and Axles Analysis Summary



DRIVETRAIN

Reduction Box, Rear Hubs and Axles Testing / Validation

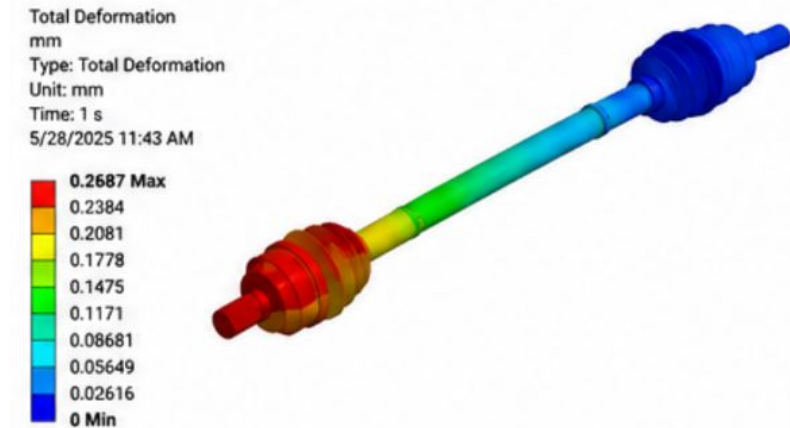
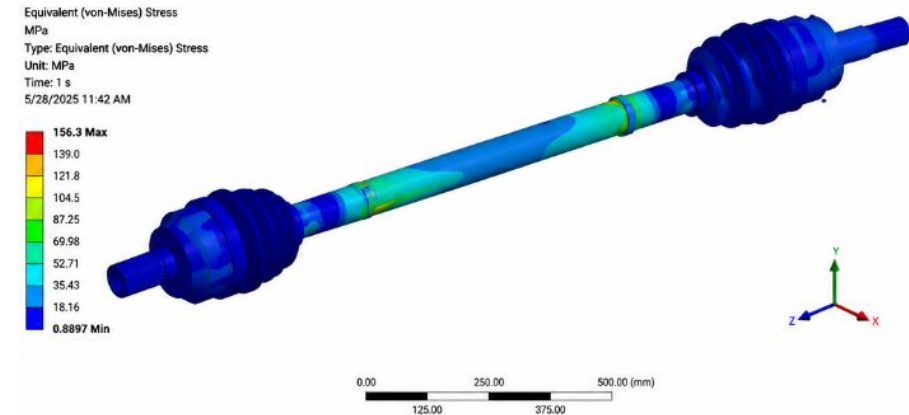
Reduction Box

- FEA-tested drive shaft lock under engagement and disengagement under load.
- Yamaha Grizzly manufacturer data for front differential.
 - Peak torque specs
 - Load data
 - Design Reviews

Axles

- Evaluated CV axles and joints for stress, deformation, and potential failure under peak loads.

Reduction Box, Rear Hubs and Axles Correlation of Analysis and Testing



DRIVETRAIN

Reduction Box, Rear Hubs and Axles Manufacturability

Reduction Box:

- Repurposed an off-the-shelf Yamaha Grizzly 660 differential, reducing manufacturing scope.
- Used commercial sprockets for easy final-drive ratio changes.

Axles:

- Selected standard ATV-grade CV axles to eliminate custom fabrication and heat treatment.
- Extended CV length with researched method

Reduction Box, Rear Hubs and Axles Serviceability



DRIVETRAIN

- **Thorough documentation and simulation** used to evaluate CVT thermal behavior
- Analyzed **heat generation and temperature distribution** under representative operating conditions
- Results were used to inform **CVT cooling and packaging decisions**

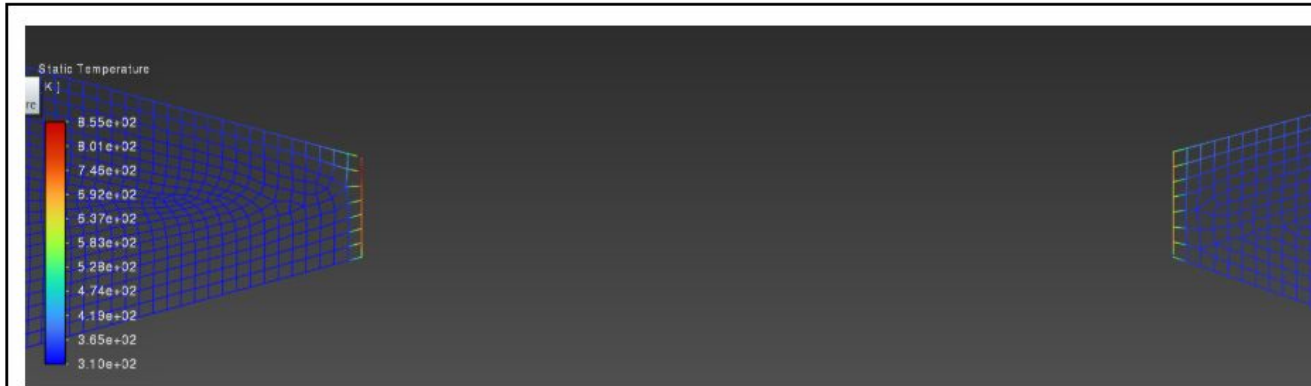
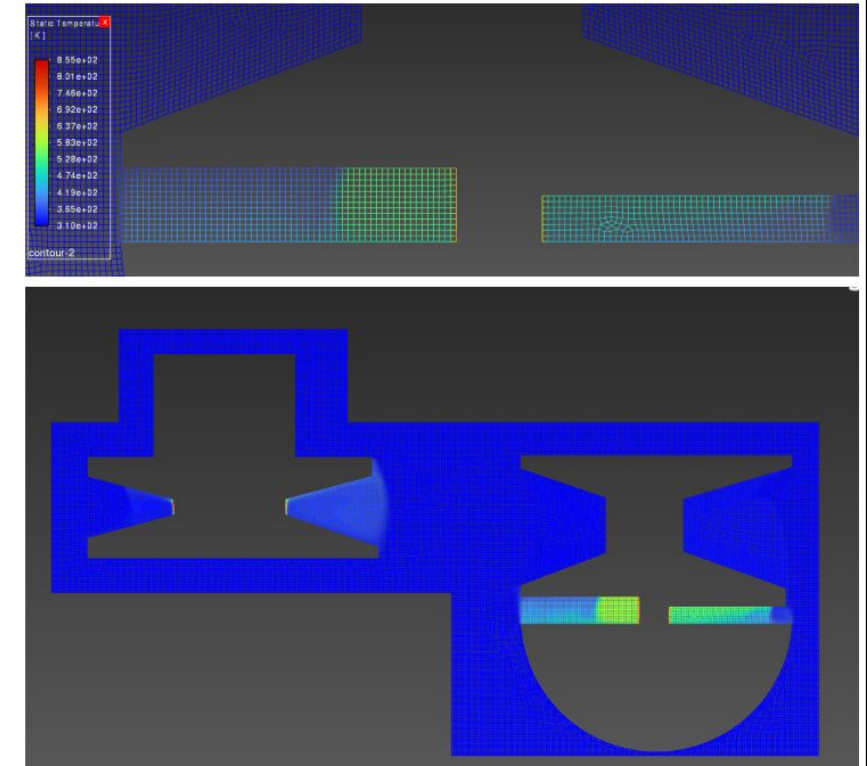


Figure 1: Wall thickness 3 mm, low rotational speed



Figures 2 & 3: Wall thickness 3mm, high rotational speed

2026 Baja SAE Design Review Briefing (DRB)

School Name

University of Texas at Austin

2026 Car Number(s)

185

SUSPENSION, STEERING AND BRAKES

Requirements

Compliance:

- Provide minimum 8" wheel travel with margin for durability
- Maintain positive Ackermann geometry through full steering lock, no binding
- Deliver 4-wheel hydraulic braking capable of locking all wheels on demand

Integration:

- Suspension geometry must clear chassis and roll cage members through full travel
- Steering column must interface cleanly with pedal box and driver ergonomics envelope
- Hub/upright design must accommodate Yamaha Grizzly 660 CV and differential interface

Manufacturability:

- Custom 6061 hubs/uprights sized for CNC or in-house fabrication, not outsourced
- Minimize custom components by integrating OEM Grizzly 660 hubs and CVs where possible
- Heim joints and weld nuts sized to standard, sourceable hardware (no custom fasteners)

Weight:

- Suspension and steering components sized to survive worst-case 3–7g bump loading without excess material margin

Research and Development

Performance:

- Built a 30-sheet, ~7,600-formula Vehicle Dynamics Calculator to validate geometry and loads from first principles as no historical data existed to design against.
- Simulated suspension kinematics in Lotus (Suspension Analysis) to validate camber, toe, and roll center behavior through full wheel travel before committing to final geometry.
- Steering column stress-analyzed using Shigley's methodology against the Narrco rack interface to confirm survival under worst-case input loads.

Manufacturing:

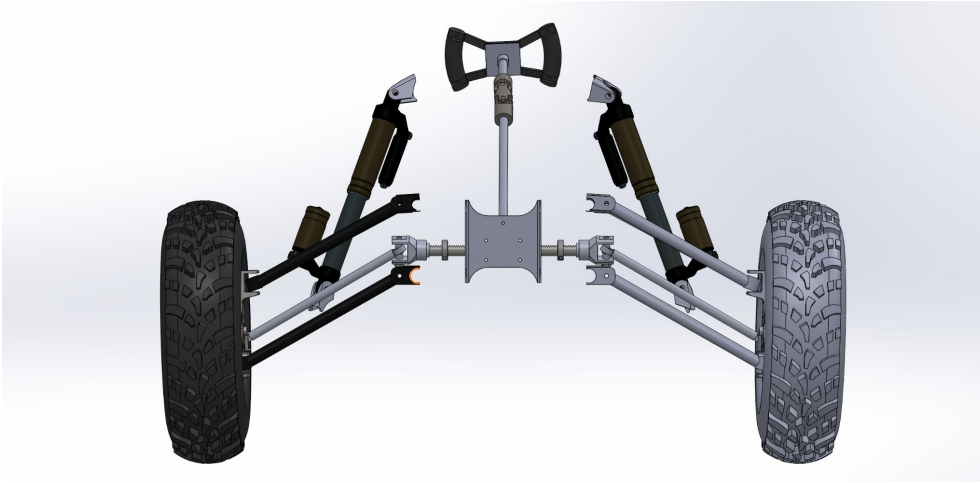
- Leveraged OEM Yamaha Grizzly 660 hubs and CVs to reduce fabrication risk on safety-critical rotating components as a first-year team.
- CV axle length extended from ~10" to ~20" cup-to-cup via extending shafts rather than custom-designing axles from scratch.

Compliance:

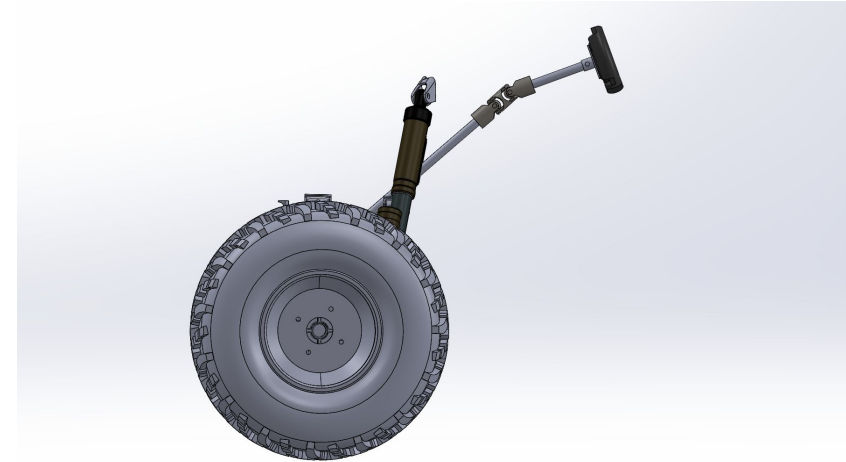
- Heim joint sizing re-verified at 1/2" Grade 8 in double shear under the 3–6g bump case; 3/8" was rejected as insufficient
- Calculator and analysis now serve as the load/geometry baseline for future LBR design cycles

SUSPENSION, STEERING AND BRAKES

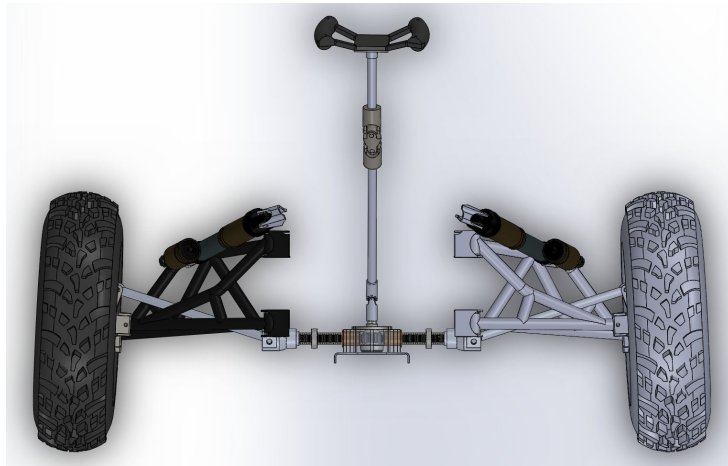
Front Suspension and Steering CAD – Front View



Front Suspension and Steering CAD – Left Side View



Front Suspension and Steering CAD – Top View



Front Suspension and Steering CAD – Isometric View

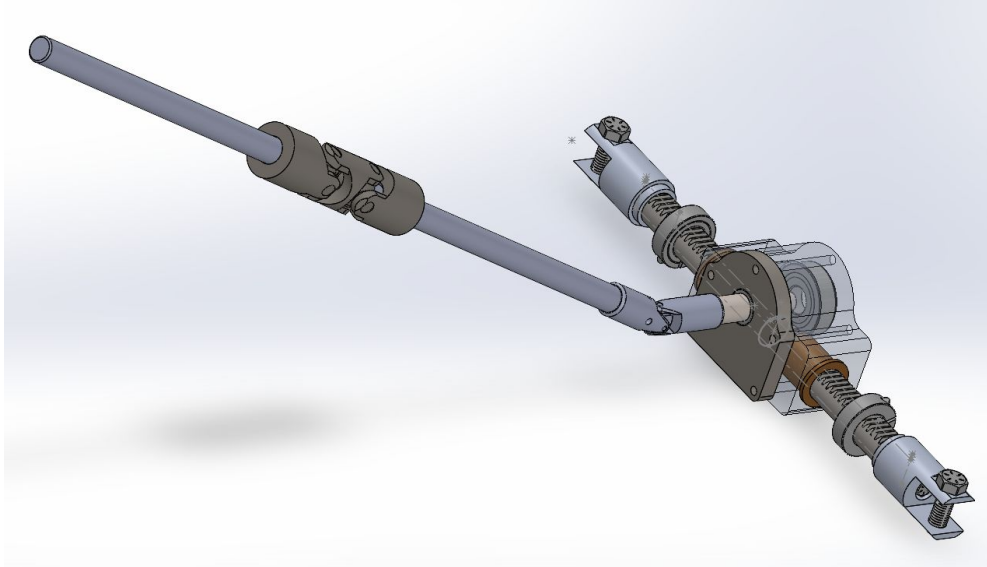


SUSPENSION, STEERING AND BRAKES

Front Suspension and Steering Analysis Summary

- **Original Steering Setup**

- Rack length - 18.75 inches
- Anti Ackerman steering chosen for low speeds
- Double U-joint connection
- Custom Tie rod Extension
- Subject to high torsion
- Expensive



Front Suspension and Steering Analysis Summary

- **Redesign Steering Setup:**

- Smaller rack length and side chassis tabs (instead of front tabs) to allocate space for ergonomics parts (pedal box, brake lines, etc)
- Ackerman steering chosen for ease of manufacturing
- No U-joint setup for ease of assembly and to reduce column support weight and complexity
- Rotated Narrco rack stub to allow for upright steering
- Went with redesign for ease of assembly and manufacturing as a first year team, and to learn from rack design for future years.



SUSPENSION, STEERING AND BRAKES

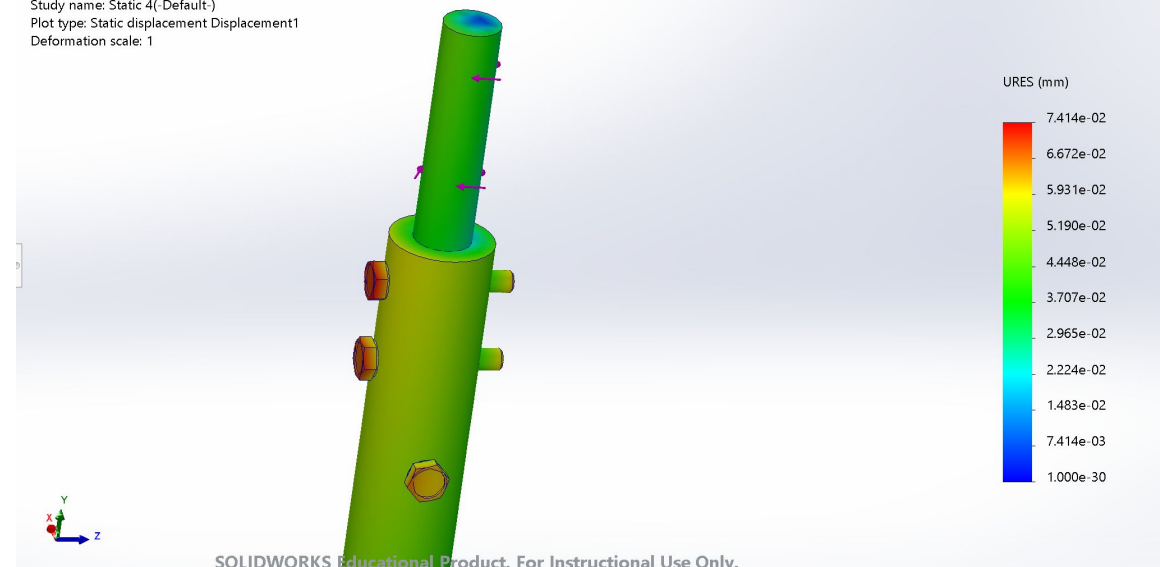
Front Suspension and Steering Testing / Validation

- Front suspension analyzed for 3g loads, majority on lower control arm, major stress point at weld point for weld cup with minimal deflection overall, giving confidence in the overall design for manufacturing
- Steering column analyzed in the following order:
 1. Stress
 - a. Stress concentration factors
 - b. Transverse and torsional shear stress
 - c. Bending stress
 - d. Von Mises stress
 - e. Pin bearing stress
 2. Shear yield strength
 3. Angle of twist
 4. Factor of safety
 - a. Torsional
 - b. Bending
 - c. Yield
 - d. Pin shear
 - e. Pin bearing
 - f. Pin tear-out
 5. Column buckling
 6. Critical unit load

Front Suspension and Steering Correlation of Analysis and Testing

- Factor of safety results:
 - Torsional $\rightarrow 1.9$
 - Bending $\rightarrow 6.9$
 - Yield $\rightarrow 1.7$
 - Pin shear $\rightarrow 3.4$
 - Pin bearing $\rightarrow 77.6$
 - Pin tear-out $\rightarrow 14.0$
- Angle of twist = 0.46 deg

Model name: steering column assembly
Study name: Static 4(- Default-)
Plot type: Static displacement Displacement1
Deformation scale: 1



SUSPENSION, STEERING AND BRAKES

Front Suspension and Steering Data Collection for Analysis

Front Spring and Damper Choice

- 125lb/in Spring rates chosen using hand calcs for target ride frequency and motion ratios. Single rate suspension design for simplicity with 7lb/in helper spring for ride height adjustment
- Fox coilover dampers chosen for cost-effectiveness, durability, travel, and adjustability.



Front Suspension and Steering Analysis Summary

Front Suspension Geometry Optimization

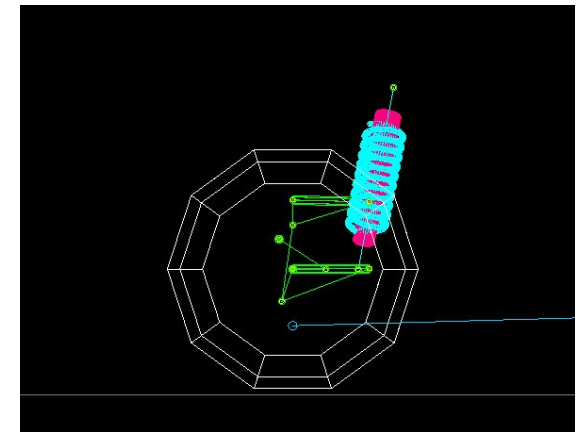
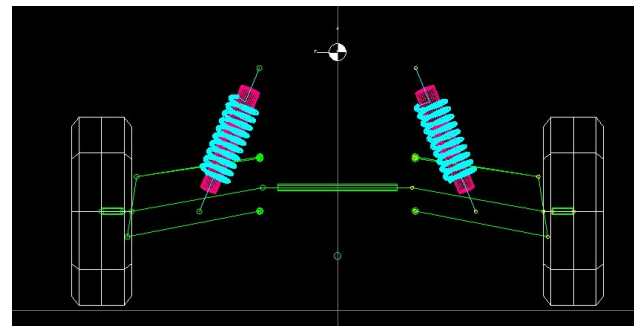
- Suspension hardpoints were iterated through Lotus Shark
- Dynamic characteristics like toe, camber, caster, KPI adjusted within acceptable ranges for offroad vehicles.

Steering Setup

- Ackermann steering setup chosen to eliminate tire scrub and low speed stability.

Front Suspension Analysis

- FEA -
- Geometry - Clearance of the lower control arms around the shocks analyzed and validated through CAD



SUSPENSION, STEERING AND BRAKES

Front Suspension and Steering Testing / Validation

Suspension Testing and Validation

- Proper motion of front control arms through bump and droop were as expected. No interferences between shock and control arms
- Validate chosen spring rates using dynamic testing
 - Identify understeer or oversteer behaviour
 - Identify over dampened or stiff coilovers
- Validate camber and toe change through travel
 - Using stands and jigs to simulate suspension load, measure actual values vs expected values



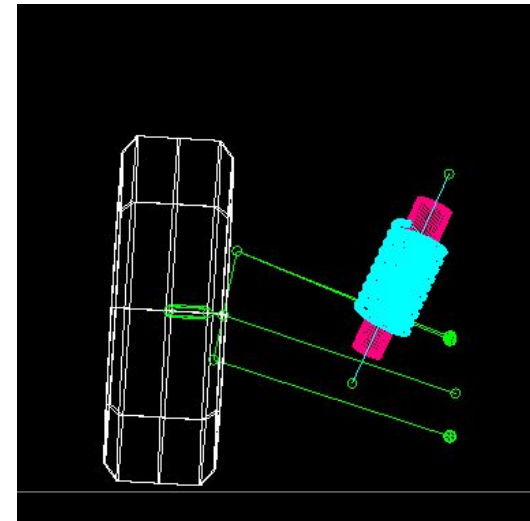
Front Suspension and Steering Correlation of Analysis and Testing

Spring and Shock Tuning

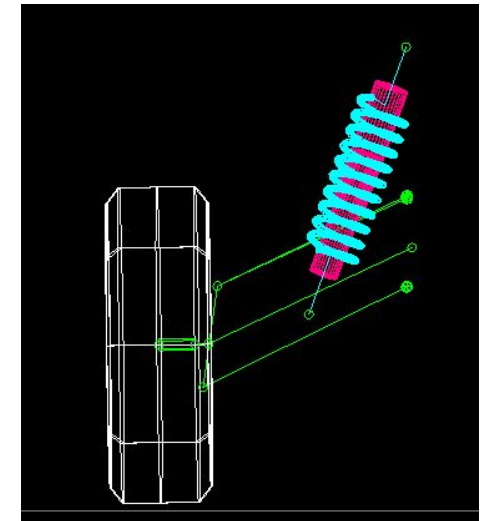
- Helper spring preload adjusted to match expected ride heights
- Damper settings set to driver preferences and based on variable conditions
- Limit straps used to prevent spring chatter and bottoming out the shocks

Camber and Toe Changes (bump steer)

- Initial camber and toe changes are as expected



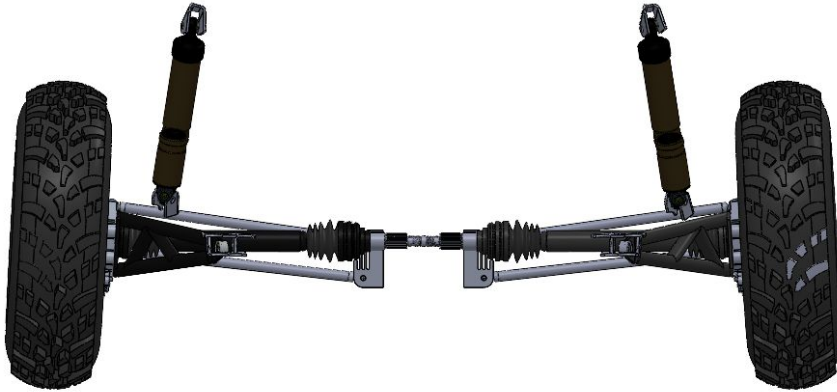
Front in 8" bump



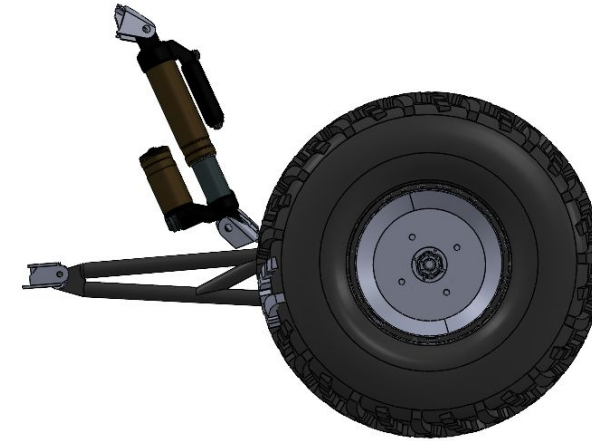
Front in 4" droop

SUSPENSION, STEERING AND BRAKES

Rear Suspension CAD OLD Design– Front View



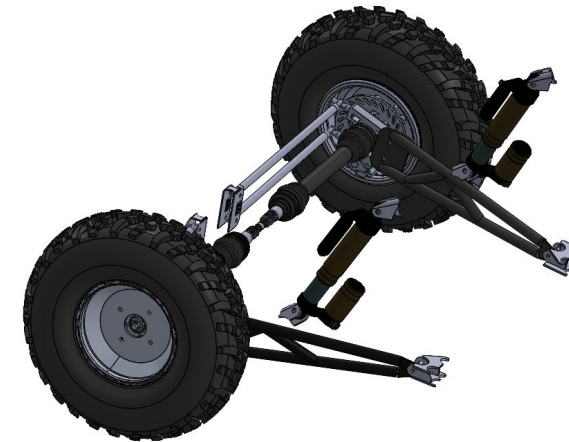
Rear Suspension CAD OLD Design – Left Side View



Rear Suspension CAD OLD Design – Top View

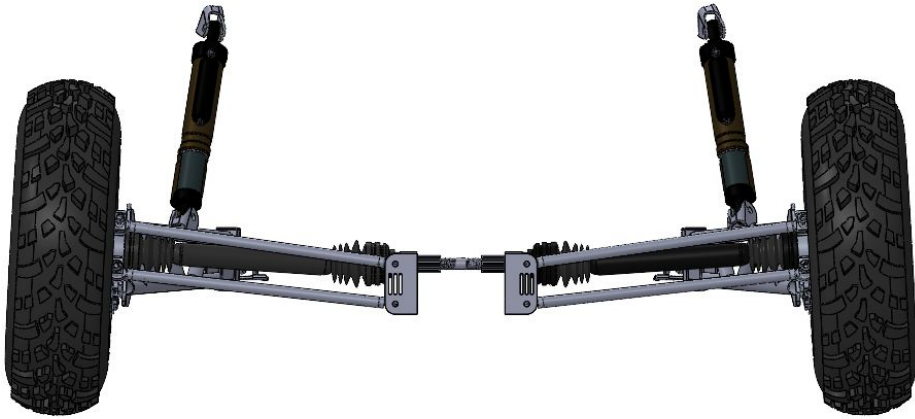


Rear Suspension CAD OLD Design – Isometric View

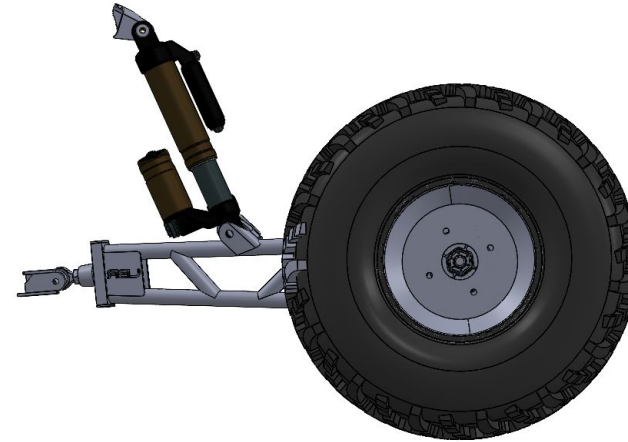


SUSPENSION, STEERING AND BRAKES

Rear Suspension CAD – Back View



Rear Suspension CAD – Left Side View



Rear Suspension CAD – Top View



Rear Suspension CAD – Isometric View



SUSPENSION, STEERING AND BRAKES

Rear Suspension Data Collection for Analysis

Suspension Geometry Selection

- Trailing arms chosen in consideration of: simplicity, manufacturability, wheel travel, and toe change.
- Heim joints were oversized (to prevent bending failure) and put on each joint to allow adjustability allowing for easy suspension adjustment.

Rear Spring and Damper Choice

- 210lb/in Spring rates chosen using hand calcs for target ride frequency and motion ratios. Single rate suspension design for simplicity with 7lb/in helper spring for ride height adjustment
- Dampers chosen for cost-effectiveness, durability, travel, and adjustability.



Rear Suspension Analysis Summary

Rear Suspension Geometry Optimization

- Suspension hardpoints were iterated through Lotus Shark program
- Dynamic characteristics such as toe and camber adjusted within acceptable ranges for offroad vehicles.

Rear Suspension Analysis

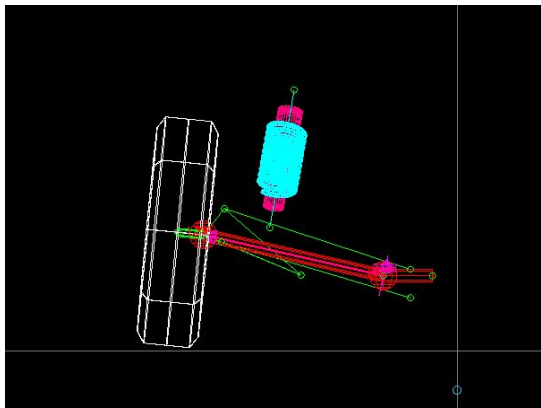
- Geometry - Compatibility with CV Axles and shock clearance was analyzed and validated through CAD
- Trailing arm chassis board mounting was switched from spherical bearing to heim joint.
 - Allows for increased adjustability to account for errors from chassis manufacturing tolerances
 - The heim joint was oversized to ensure it could survive bending loads, a typical failure point on heim joints.

SUSPENSION, STEERING AND BRAKES

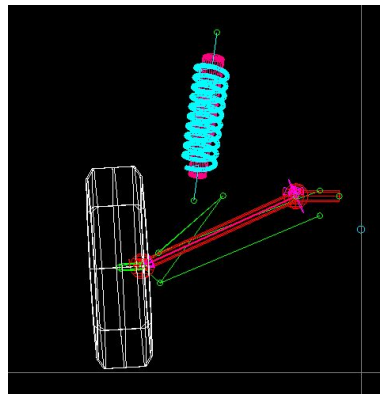
Rear Suspension Testing / Validation

Suspension Testing and Validation

- Proper motion of rear trailing through bump and droop were as expected. No interferences between shock and chassis.
- Validate chosen spring rates using dynamic testing
 - Identify over dampened or stiff coilovers
- Validate camber and toe change through travel
 - Using stands and jigs to simulate suspension load, measure actual values vs expected values
- Extended CV axles ensure proper range of motion and track width



Rear 8" bump



Rear 4" droop

Rear Suspension Correlation of Analysis and Testing

Spring and Shock Tuning

- Helper spring preload adjusted to match expected ride heights
- Damper settings set to driver preferences and based on variable conditions
- Limit straps used to prevent spring chatter and bottoming out the shocks

Camber and Toe Changes

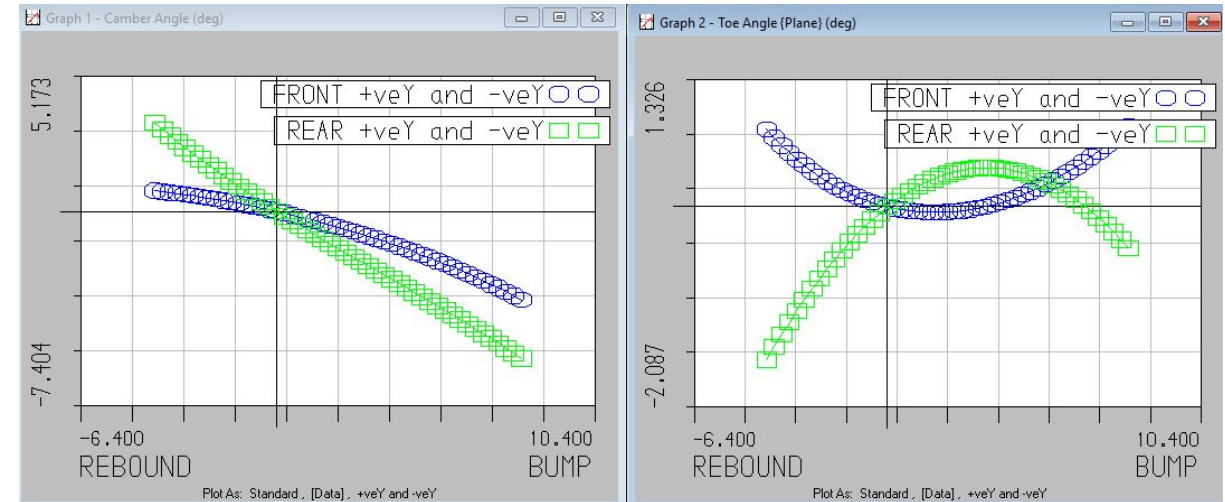
- Toe links adjusted to desired range and properly constrain motion
- Camber change over 12" suspension travel is within range

SUSPENSION, STEERING AND BRAKES

Overall Suspension

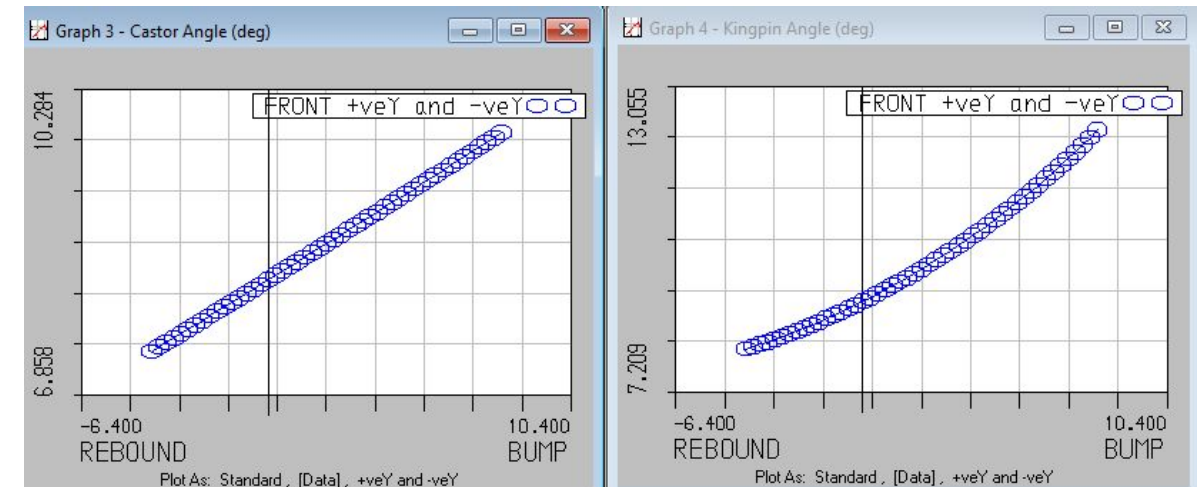
Expected Values

- Front
 - 0.787° to -3.373° camber angle
 - Toe Change $\Delta < 1^{\circ}$ over full travel
 - 8.1° static caster
 - 8.952° static KPI
- Rear
 - 3.376° to -5.608° camber angle
 - Toe Change $\Delta < 1^{\circ}$ over useable bump range
- Overall
 - Roll Centre Height Δ Rear - Front 2" through travel



Camber Graph

Toe Graph



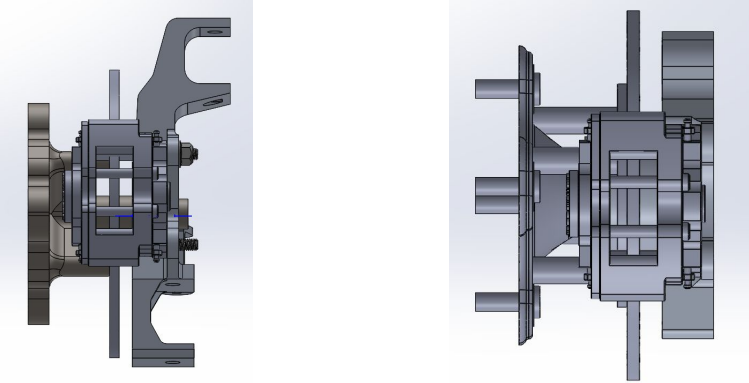
Castor Graph

KPI Graph

SUSPENSION, STEERING AND BRAKES

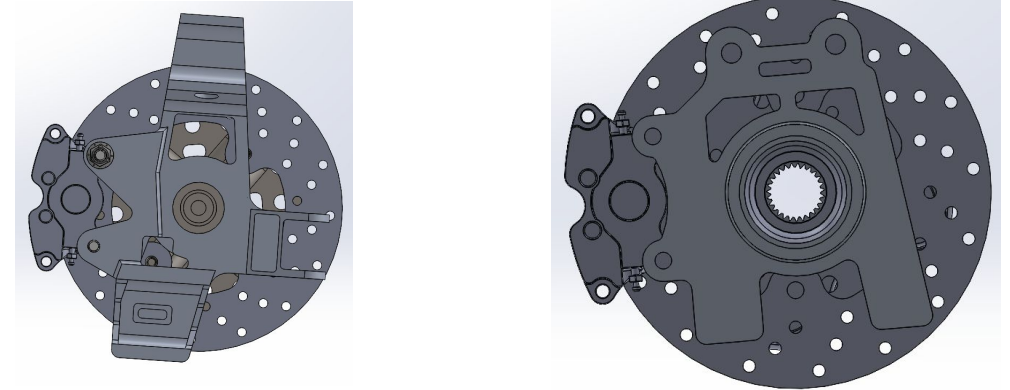
Brakes CAD – Front View

Front / Rear



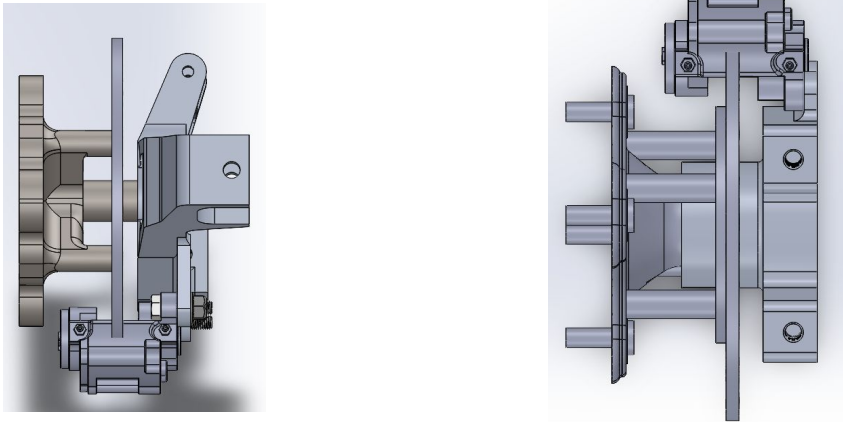
Brakes CAD – Left Side View

Front / Rear



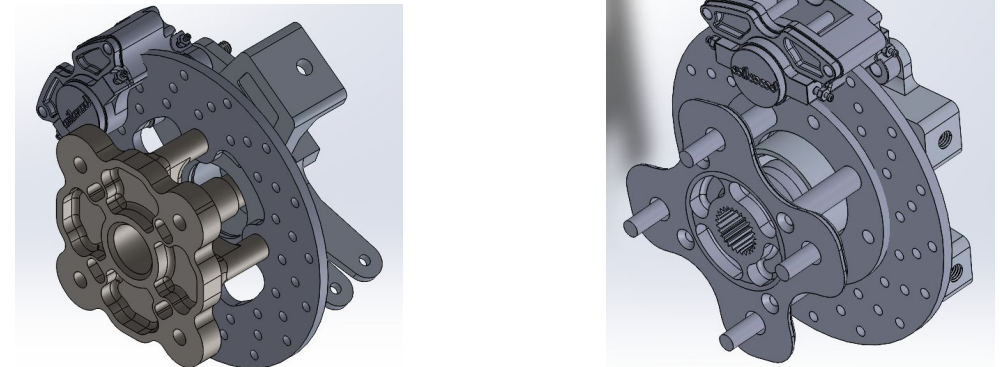
Brakes CAD – Top View

Front / Rear



Brakes CAD – Isometric View

Front / Rear



SUSPENSION, STEERING AND BRAKES

Brakes Data Collection for Analysis

Vehicle Parameters

- Vehicle & Driver mass: 660 lbs
- Wheelbase: 69.5 in
- CG height / position: 29.4 in
- Static front-rear weight distribution: 60/40
- Tire radius: 11.5 in
- Tire-road friction coefficient: 0.8

Brake Hardware

- Pedal ratio: 5
- Master cylinder bore: 0.625 in
- Caliper piston area: 1.23 in²
- Rotor effective radius: 3.1 in
- Pad friction coefficient: 0.3 (~lower limit)

Driver Input

- Pedal force: 110 lb

Brakes Analysis Summary

- Target deceleration: 25.7 ft/s²
- Required pedal force: 110 lb
- Weight transfer: 223.5 lb
- Front brake bias: 75%
- Rear brake bias: 25%
- Front line pressure: 1344.5 psi
- Rear line pressure: 448.2 psi
- Front brake torque: 3076.0 in*lb
- Rear brake torque: 1025.3 in*lb
- Predicted stopping distance from 25 mph: 26 ft



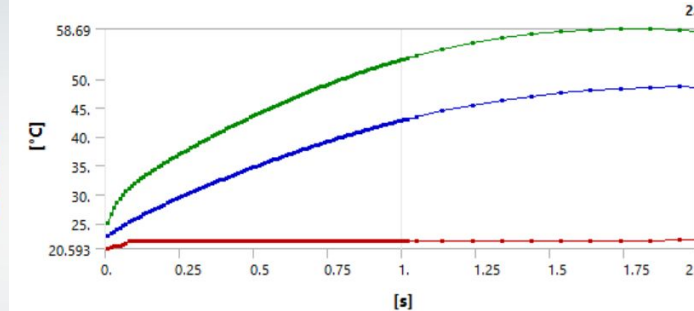
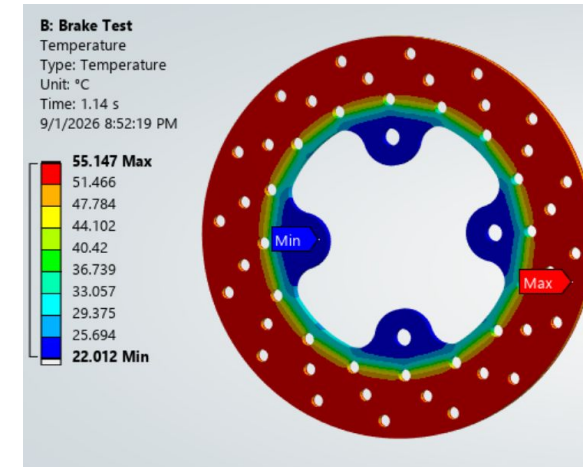
SUSPENSION, STEERING AND BRAKES

Brakes Testing - Competition Brake Test

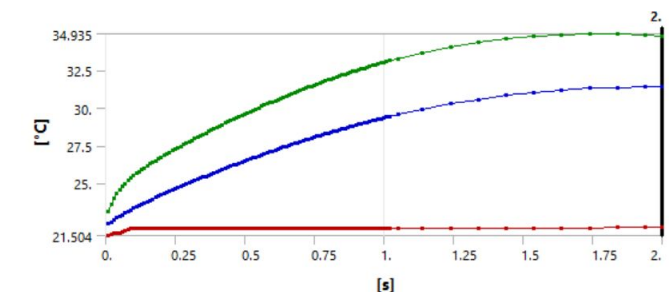
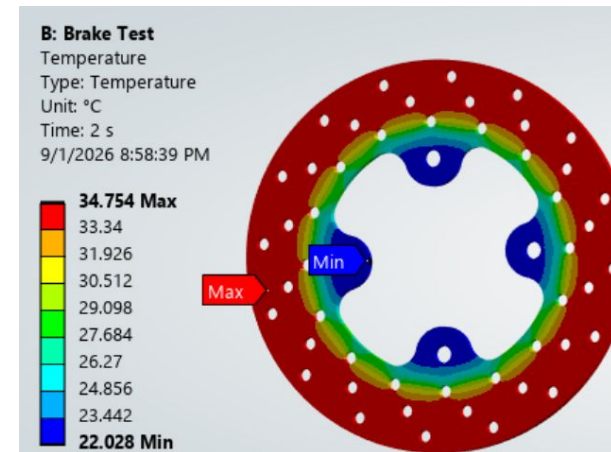
Analyzed:

Brake Rotor Temperature vs Time

- Calculations assumed a conservative braking condition with braking from 25 mph to 0 mph with ambient temperature brake discs
- Results show that wheels will lock assuming input force of 110 lb (brake pad coefficient of friction at these temperatures are roughly .3)



Front



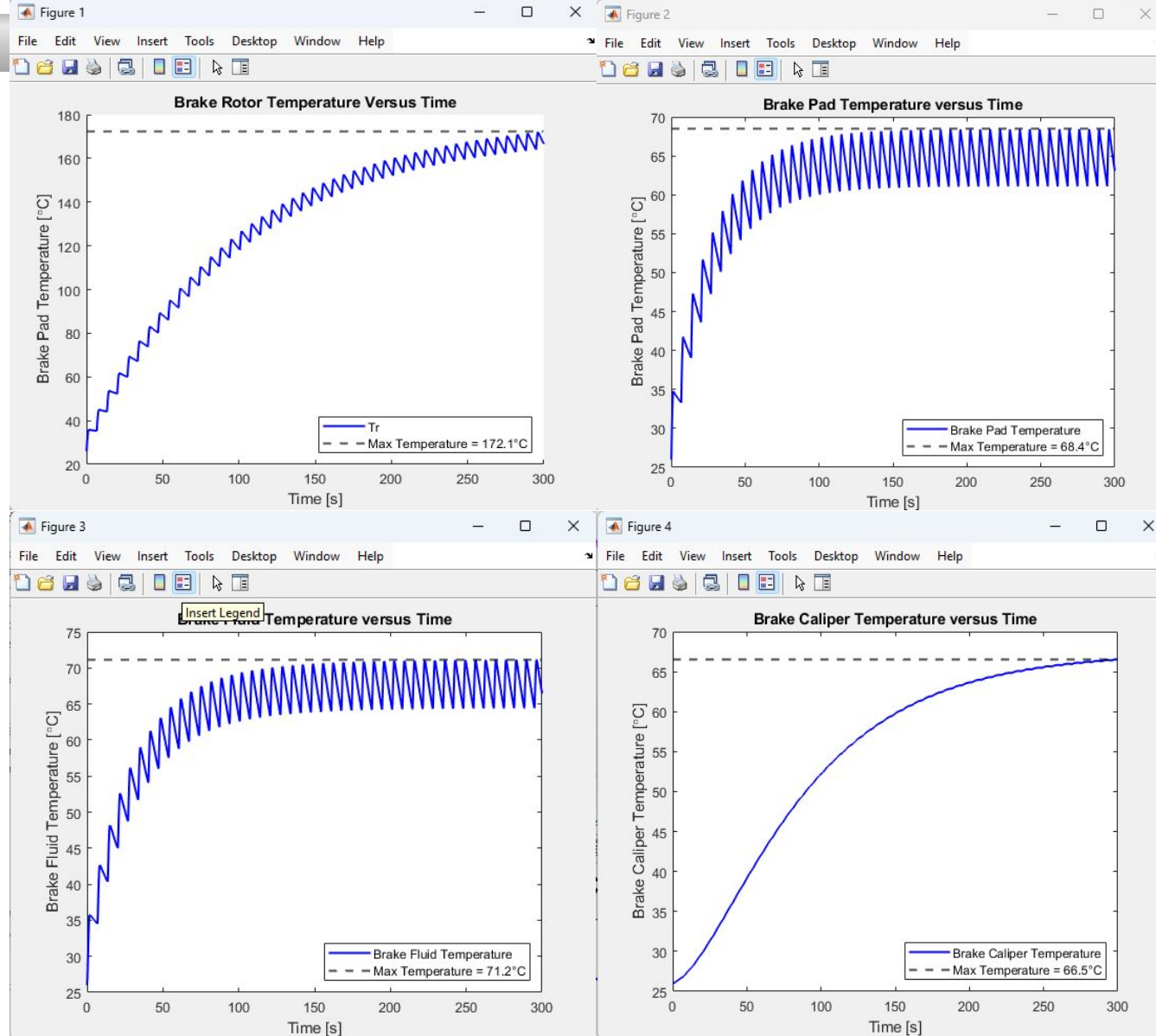
Rear

SUSPENSION, STEERING AND BRAKES

Brakes Testing - Temperature

Analyzed:

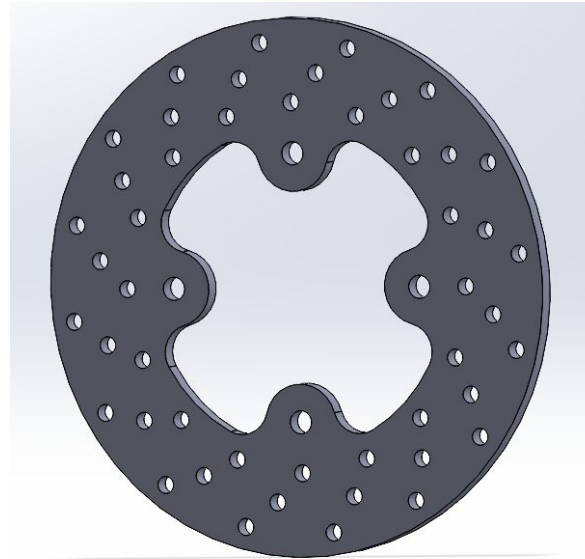
1. Brake Rotor Temperature vs Time
 2. Brake Pad Temperature vs Time
 3. Brake Fluid Temperature vs Time
 4. Brake Caliper Temperature vs Time
- Calculations assumed a conservative braking condition with repeating accelerating and braking from 35 mph to 10 mph for 300 seconds
 - Rotor survives warping at max temperature
 - Braking fluid does not boil off at max temperature



SUSPENSION, STEERING AND BRAKES

Brake Rotor Manufacturability

- Custom rotors were selected because we encountered mounting hole location issues with our OEM hubs
- Goal was to maintain structural material around mounting holes to prevent failure
- Allowed us to add material cutouts, reducing overall rotor weight by 1 lb

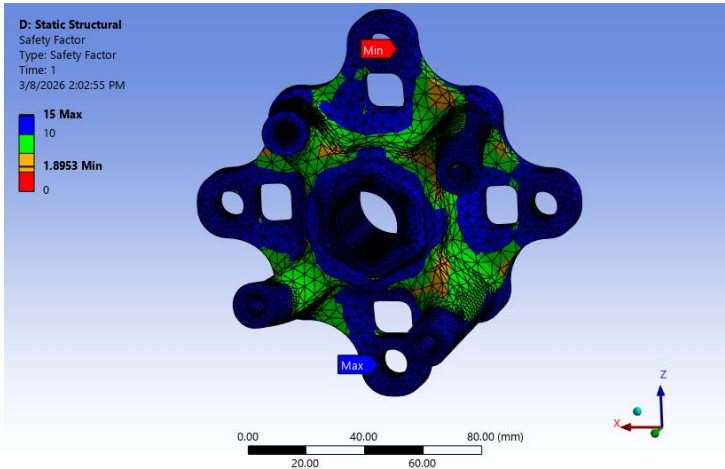
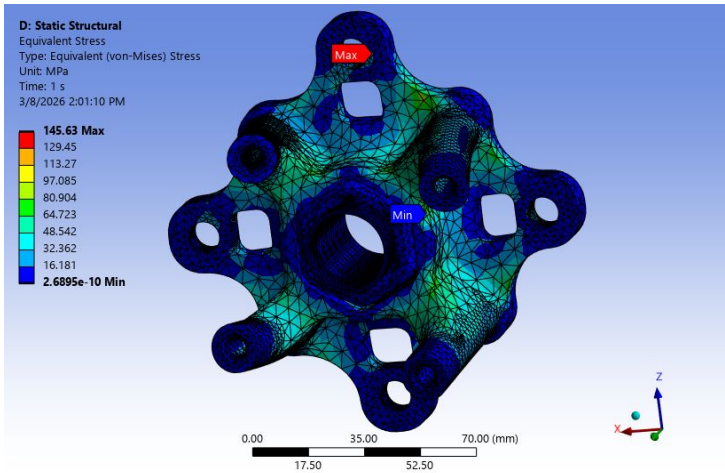


Selected for Manufacture

SUSPENSION, STEERING AND BRAKES

Rear Hub: Original Design Maximum Load Case (Bump + Steering)

- Original Rear Hub design meant to work with Grizzly 660 Spline pattern and rims
- Utilized hexagonal spline plates cut from 6061 Aluminum welded into a stack up creating the slime profile to transfer torque from differential into the wheel.
- Design was evaluated and manufactured, however due to time and budget constraints, development was held off to work with a modified OEM hub for simplicity and verified compatibility under heavy load from the powertrain subsystem.



Sim 5: Jump + Steering (Worst Case)				
Forces	Amount	Units	Direction	Where it acts
Jump Vertical Load	2668.93	Newtons	+Y	Wheel Center
Longitudinal Friction (equal split)	1509.775	Newtons	-X	Wheel Center / tire contact
Lateral Friction (equal split)	1509.775	Newtons	+Z / lateral	Wheel Center / tire contact
Hub Torque	441.005	N·m	About hub axis	Hub center / brake interface

SUSPENSION, STEERING AND BRAKES

Rear Hub Validation

- **Load Case Force Modeling**

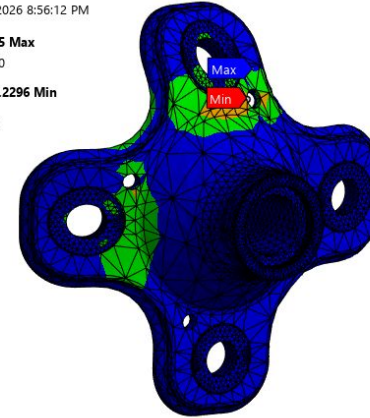
- Utilized derived equations of motion, creating free-body diagrams to determine input force from tires, to brake rotors, to rear hub
- Input parameters:
 - Vehicle mass (with driver), weight distribution (Front = 40%, Rear = 60%), wheelbase, track width, tire radius, bump/jump vertical loads (5g), material properties (structural steel)
- Returned values:
 - Dynamic wheel vertical load, longitudinal/lateral friction forces, resultant hub torque about the axle
- Governing (worst-case) load case: Jump + Steering with simultaneous longitudinal and lateral friction at the tire contact patch

- **All Forces put into ANSYS Static Structural**

Rear Hub Analysis Summary

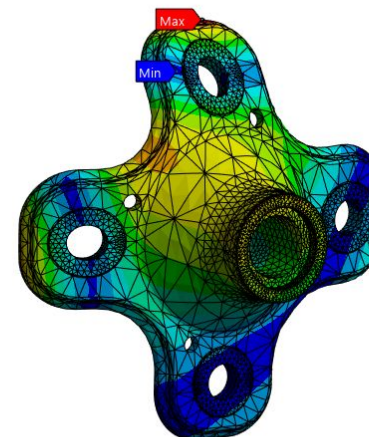
B: Sim 5: Jump + Steering
Safety Factor
Type: Safety Factor
Time: 1
8/31/2026 8:56:12 PM

15 Max
10
3.2296 Min
0



B: Sim 5: Jump + Steering
Total Deformation
Type: Total Deformation
Unit: mm
Time: 1 s
8/31/2026 8:57:31 PM

0.025854 Max
0.023003
0.020151
0.017299
0.014447
0.011595
0.0087435
0.0058916
0.0030398
0.000188 Min

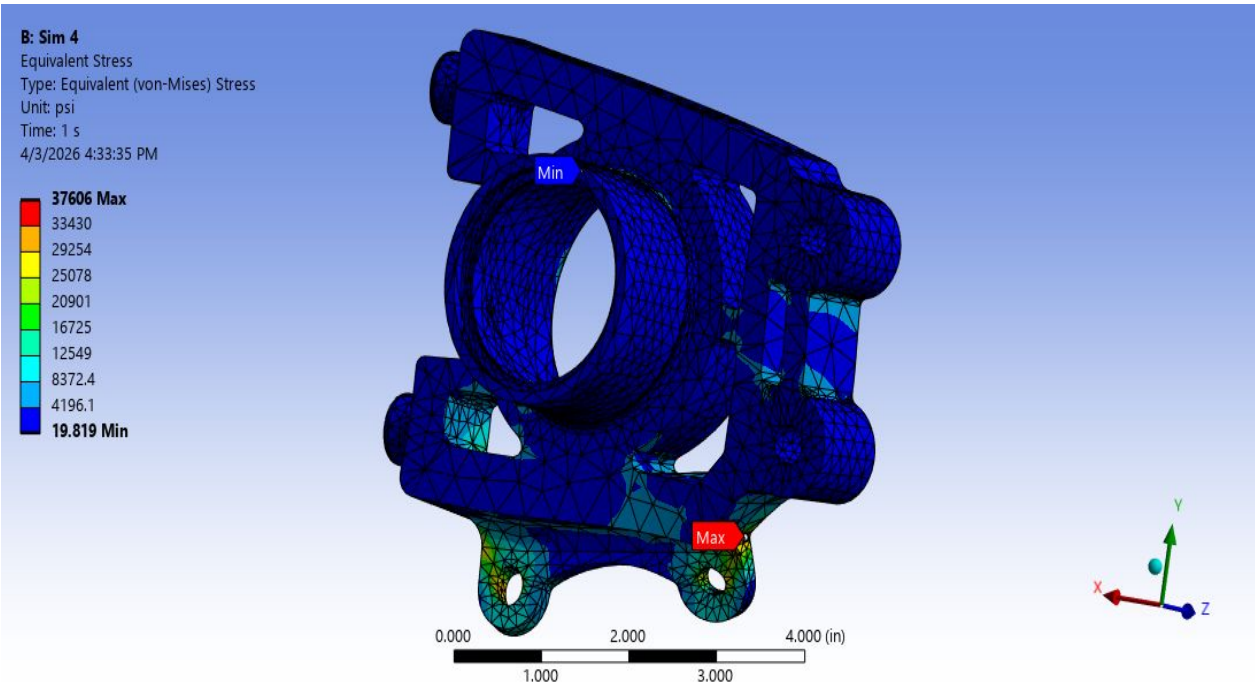
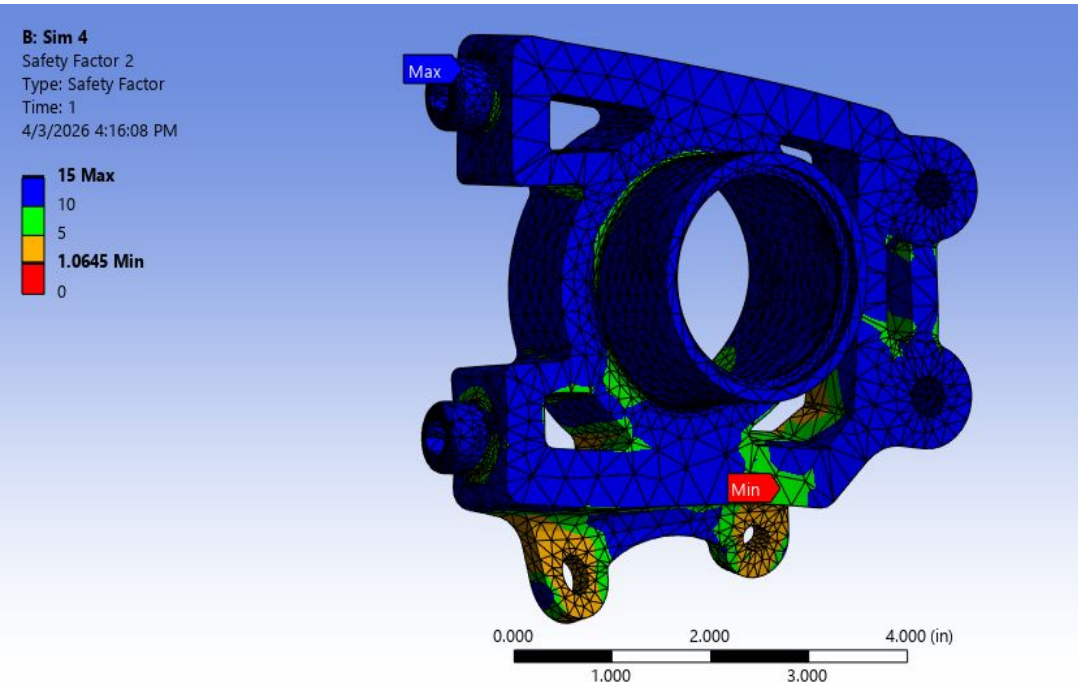


- **Jump Vertical Load (5g)**
 - 4003.395 Newtons in the Y+ applied at the hub center
- **Longitudinal Friction (equal split)**
 - 2265 Newtons in the -X applied at the hub center
- **Lateral Friction (equal split)**
 - 2265 Newtons in the +Z/Lateral applied at the wheel center
- **Hub Torque**
 - 661.38 Newtons * Meters applied counter clockwise along the hub axis
- **Ensure FoS > 3.2296**

SUSPENSION, STEERING AND BRAKES

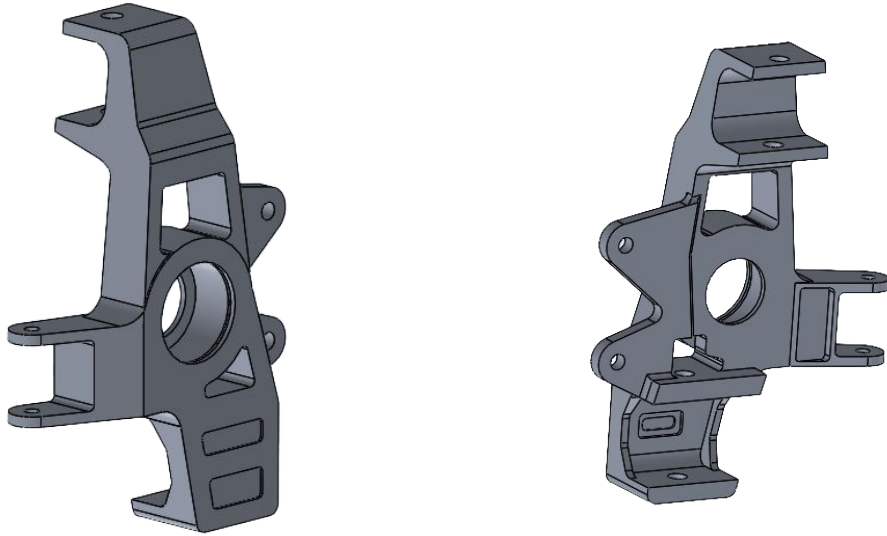
Rear Upright: Steering while accelerating while going over a bump (Worst Case)

Load Case	Scenario	Vertical Load Fz	Longitudinal Load Fx	Lateral Load Fy	Units
Sim 4	Steering while accelerating while going over a bump (Worst Case)	2649.859	1405.300	1405.300	N



SUSPENSION, STEERING AND BRAKES

Front Upright Isometric View



Right Front Upright:

- 6061-T6 Aluminum for strength and weight optimization
- H7 Bearing sleeve tolerance
- Total weight: 1.85 lbs

Front Wheel Package - Isometric, Cross-Sectional, Exploded



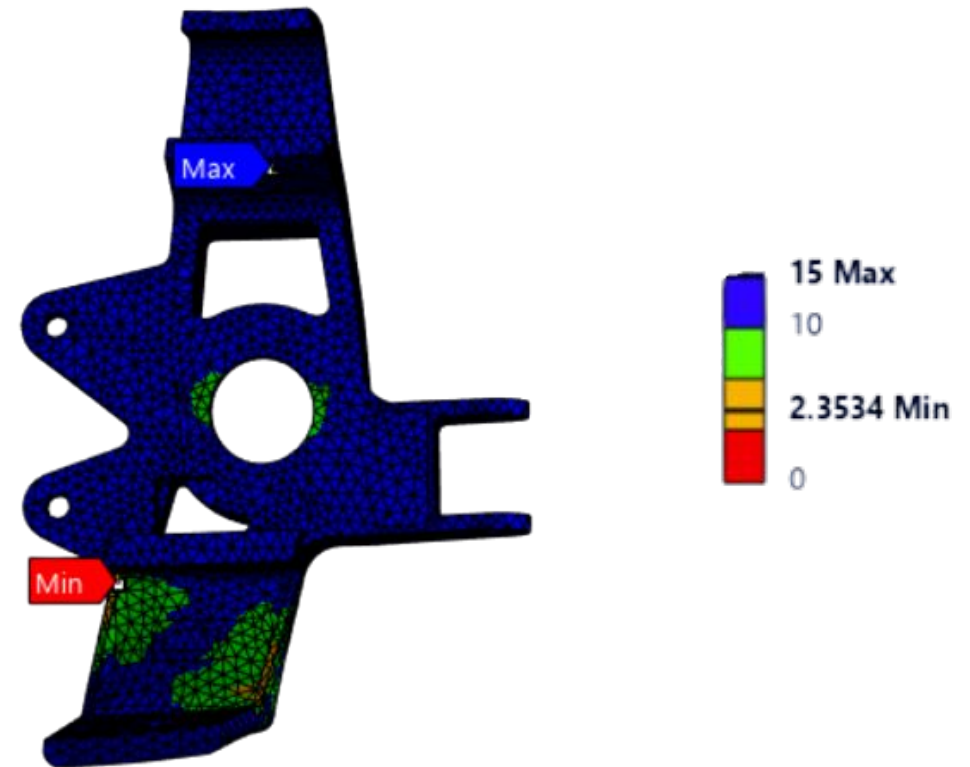
SUSPENSION, STEERING AND BRAKES

Front Upright Validation

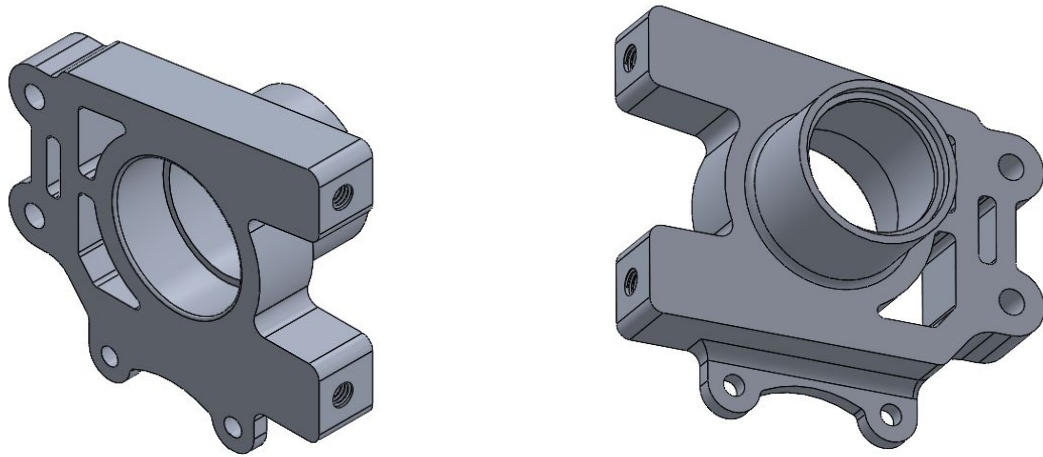
- **Braking @ 15° decline with Steering: 2.35 FOS**
 - Fixture at upper and lower mounting arm
 - Calculated load force applied to the bearing sleeve from tire contact patch
- **7G Vertical, 3G lateral, 2.15G Braking: 0.84 FoS**
 - Fixture at upper and lower mounting arm
 - Calculated load forces applied to bearing sleeve from tire contact patch

Front Upright FEA Validation:

Braking @ 15° decline with Steering:



Rear Upright - Isometric View



Rear Upright

- 6061-T6 Aluminum Upright CNC milled out-of-house. Internal bore refined in-house with manual lathe. Bolt holes tapped and threaded in-house with manual mill.
- H7 Interference fit between wheel bearing and upright.
- Total Weight: 1.61 lbs

Rear Wheel Package - Isometric, Cross Sectional, Exploded

